

Review

Microencapsulated phase change materials for enhanced thermal energy storage performance in construction materials: A critical review

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ABSTRACT

The potential of phase change materials (PCM) as a thermal energy storage medium in buildings has been widely discussed. However, the possible leakage of melted PCM into construction material matrix could have deleterious effects on some of the intrinsic properties of these materials. To prevent this, PCM is microencapsulated in micron size shells to form microencapsulated phase change materials (MPCM). Numerous studies in the literature, including reviews, have shown that MPCM can enhance the thermal performance of construction materials and reduce operational carbon emissions associated with frequent heating and cooling of buildings. However, there is limited information on the effects of the shell material encapsulating the PCM on the thermal performance of PCM as well as its effect on the inherent properties of cementitious composites and gypsum boards. This information is extremely useful for better understanding of how the known properties of construction material will be affected by the incorporation of MPCM into its matrix. In this study, we critically analyzed how MPCM affects the fresh and hardened properties of construction materials. More also, general overview of PCM, microcapsules, production process of MPCM, and various thermal enhancement methods for MPCM were also rigorously discussed and evaluated. The findings from this study show that biobased PCM can be an environmentally friendly alternative replacement for paraffins and MPCM can improve thermal performance of construction materials. The low-cost advantage of inorganic PCM over organic PCM makes it a potential candidate for economical thermal energy storage solutions. Furthermore, MPCM typically lowers the mechanical properties of building materials, with a considerable loss reported in MPCM with organic shells as opposed to inorganic microcapsules. The workability of cementitious composite decreases with increasing MPCM content, and the shell material and morphology are significant contributing factors. The highlighted knowledge gaps for future research to support MPCM scalability and large-scale applications were critically reviewed. The findings from this review provide invaluable insights for researchers, engineers, and key stakeholders in the construction industry on the recent developments and huge potential of MPCM for building thermal management.

1. Introduction

Amidst the looming energy crisis and climate emergency, enhancing the energy efficiency of buildings and developing cost-effective energy storage solutions represent a crucial step towards achieving a sustainable future. Thermal energy storage (TES) employing phase change materials (PCM) has emerged as an attractive option for improving building energy efficiency, attracting considerable interest from

academics, engineers, and policymakers [1–4]. PCM can store thermal energy by undergoing a phase transition from solid to liquid or vice versa and release it at peak energy demand, allowing buildings to lessen their dependency on fossil fuel-based energy sources [5]. This consequently allows for the adjustment of a building's diurnal ambient temperature. Incorporating PCM into building components like walls, floors, and ceilings can help shift energy demand to off-peak hours, flatten internal temperatures, and use free cooling at night to reduce peak

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cooling load during the day [6–8].

The intrinsic property of PCM to change to liquid during thermal activation largely prevents their direct incorporation into construction materials. The melted PCM can affect some of the mechanical properties of construction materials and consequently reduce their TES capacity. To address this challenge, two major methods have been adopted in the literature for direct incorporation of PCM into the matrix of building materials: (i) shape stabilization and (2) microencapsulation. The shape stabilization method or form stable method entails absorbing PCM into porous host matrices such as bentonite clay [9], zeolite [10], and expanded perlite [11] to create stable composites that are then incorporated into building materials. Microencapsulation on the other hand involves encasing PCM in a protective micron size shells which are then incorporated into construction materials to improve their TES properties. In comparison to microencapsulation, the form stable approach has a higher encapsulation ratio. However, because of the host material's large porosity, the stored PCM may leak out into the surrounding matrix during melting cycles, thereby, reducing or eliminating the ability of the composite to store heat [12]. The microencapsulated PCM (MPCM) can offer a better storage of the PCM as long as the shell remains intact during thermal cycles [13]. While shape stabilized has been extensively discussed in the literature, this review is largely focused on MPCM.

MPCM can be included into building components without compromising the overall mechanical properties of the construction material. They have been applied directly to building materials as partial replacement for sand in concrete [14–16] or as percentage by weight in wallboard [17,18]. An ideal microcapsule or shell for PCM would possess certain properties to ensure the effective encapsulation of the PCM. Firstly, it should have good mechanical and thermal properties to prevent breakage during handling and usage. Secondly, it should have high thermal conductivity to facilitate the transfer of heat between the PCM and the surrounding environment. Thirdly, it should be thermally durable to withstand repeated thermal cycling without degrading or deforming. Fourthly, it should be unreactive with the PCM to prevent any chemical reactions that could alter the PCM properties. Lastly, it should prevent leakage of the PCM to ensure the stability of the microcapsule and maintain the integrity of the surrounding building materials [19].

Over the past two decades, there has been an increasing amount of review and computational studies on the development and application of PCM for TES. Some of these studies are shown in Table 1 and some observed limitation/gaps were reported. These review studies have shown that PCM can improve the indoor thermal comfort in buildings and adjust diurnal temperature. More also, the studies have investigated various aspects of PCM technology, including the preparation, characterization, and application in fields such as building energy management, solar energy storage, wind energy harvesting, textile, heat pump systems, and battery thermal management. The computational studies provided extensive research on optimizing the design and performance of PCM in buildings. Additionally, with models developed, the long-term benefits of PCM applications in buildings and its possibility to reduce operational carbon emissions from heating and cooling of buildings were predicted.

While these studies have concentrated on the preparation, optimization, and extensive uses of MPCM, there is paucity of information in these studies regarding the shell material influence on the thermal and mechanical performance of PCM as well as its effect on the intrinsic characteristics of cementitious composites and gypsum boards. This information is extremely useful for better understanding of how the known properties of construction material will be affected by the incorporation of MPCM into its matrix.

To fill the afore-mentioned knowledge gaps, this study aims to present an up-to-date information on general overview of PCM, microcapsules, production process of MPCM, and various thermal enhancement methods for MPCM. This study critically analyzes the effect of the shell material on MPCM performance and its influence on the properties of

Table 1

Some recent review and computational studies on PCM and observed knowledge gaps.

Research work	Research type	Main findings	Some limitations/gaps	Ref
Rashid et al. (2023)	review	• PCM should be selected based on climatic region • thermal stability of PCM should be guaranteed during its lifetime in construction material • cost of PCM may be one of the main roadblocks affecting wide usage	• the scope of the review did not cover analysis of effect of PCM on the intrinsic properties of building materials	[20]
Sharshir et al (2023)	review	• Bubble injections can increase PCM latent heat and reduce temperature penetration • melting temperature should be chosen close to the desired temperature and thermophysical properties of PCM should be considered when designing a system	• No evaluation was done on the effect of the shell material on the thermo-mechanical performance of the construction materials.	[21]
Chang et al. (2022)	review	• shell materials for PCM should be selected based on application requirement due to individual advantages and disadvantages of each material • MPCM enhance latent heat storage capacity of PCM and prevent their leakage • preparation methods of micro/nano capsule are rigorous which may not be industrially adopted	• analysis of the influence of shell materials on fresh and hardened properties of construction materials was not carried out	[22]
Singh Rathore et al. (2020)	review	• MPCM can improve indoor thermal comfort • proper ventilation system must be put in place to regulate heat released during crystallization of the PCM • Hybrid PCM may be suitable	• influence of shell material on thermo-mechanical performance of construction materials were not evaluated	[23]

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Table 1 (continued)

Research work	Research type	Main findings	Some limitations/gaps	Ref
		in regions with broad diurnal temperature range		
Mohseni et al. (2021)	computational	- the energy consumption of model incorporated with PCM reduced by about 71% - environmental analysis of the model showed CO₂ emissions of roughly 264 tons can be reduced in buildings with 50-year service life when integrated with PCM - The payback period would be roughly 16.6 years.	possible degradation of PCM was not considered as PCM was assumed to completely change phase diurnally during building service life	[24]
Wijesuriya et al. (2021)	computational	- Peak electricity demand can be effectively shifted by TES capacity of PCM. - While a single path approach is sufficient to record the heat transfer through PCM drywall, a parallel path method can best capture it across a building envelope with PCM pouches.	- specific heat capacity of PCM for solid and liquid phases were kept equal - the wall panels were analyzed with no insulation	[25]
Konstantinidou et al. (2018)	computational	PCM can provide improved thermal comfort and energy efficiency for buildings in the Mediterranean regions through peak load shifting	the environmental and economic analysis of using PCM in the Mediterranean regions was not analyzed	[26]
Saffari et al. 2017	computational	- proper melting temperature for PCM based on the climate is crucial during design and application - PCM with a melting temperature close to the maximum value of 26 °C is ideal for lowering annual energy usage in locations with a	The enthalpy of the PCM was assumed to be constant and possible density change during phase transition of the PCM was not considered	[27]

Table 1 (continued)

Research work	Research type	Main findings	Some limitations/gaps	Ref
		predominately cooling climate.		
		In contrast, a PCM with a lower melting temperature of 20 °C offers significant yearly energy benefits in places with a predominately warm climate.		

cementitious composites and gypsum boards. In addition, possible future applications of MPCM in construction materials were also discussed. This review is aimed at providing detailed information to researchers, engineers, and key players in the construction industry on the recent developments and huge potential of MPCM incorporation in buildings. The details provided is essential for advancing MPCM development and its incorporation into building materials, which will ultimately result in more environmentally friendly and energy-efficient buildings.

2. General overview on phase change materials

PCM have gained popularity in recent years, notably in the context of building thermal management. However, certain properties must be considered for effective application of PCM in building materials. PCM with suitable thermophysical, chemical, kinetic, environmental, and economic qualities are ideal for building applications [28]. Table 2 summarizes the specific features for an ideal PCM for building applications. These parameters govern the PCM thermal behavior, durability, cost-effectiveness, and the amount of energy that can be stored. For instance, a PCM with a high latent heat capacity may store more energy per unit mass, while a PCM with a high thermal conductivity can transmit heat quickly from and to it. To avoid any negative impacts on the environment and human health, the PCM should be chemically stable and non-toxic [28]. The low cost of PCM is essential for the broad use of PCM technology in construction applications [28,29].

In the literature, PCM has been divided into three main groups based on their chemical composition as: organic, inorganic, and eutectic

Table 2

Specific desired properties for phase change materials applicable in buildings.

Properties	Characteristics
Thermophysical	human comfort transitions temperature range, dense material with minimal volume change during transitions, phase equilibrium without separation, durability under repeated thermal cycle, high thermal conductivity and specific heat, significant latent heat of transition, enhancing heat absorption and release
Chemical	chemical stability after cycling, non-toxic, non-hazardous, non-corrosive, compatible with construction materials, complete reversibility in phase cycles.
Kinetic	rapid crystallization, does not supercool, demonstrate high rate of nucleation.
Environmental	environmentally friendly, low embodied energy, recycling potential and low environmental impact
Economic	cost-effective, plentiful, and easily accessible raw materials

compounds [30]. As seen in Fig. 1, each group has distinct benefits. While inorganic PCM are preferred for their low-cost, thermal stability and low flammability, organic PCM are also well recognized for having a high latent heat of fusion. More also, eutectic PCM have remarkable heat transfer capabilities and their potential for TES have been well discussed in [30,31].

2.1. Organic PCM

Organic PCM are a type of PCM that are made from organic compounds such as fatty acids, alcohols, and esters. They have several advantages over other types of PCM such as high latent heat capacity, low cost, non-toxicity, and biodegradability. They also have a wide range of applications in various fields such as building construction, textiles, electronics cooling, and food storage [32–34]. This group of PCM are either paraffins or non-paraffins [33]. Due to their cost, paraffins make up the majority of commercially available PCM. Their advantages include being inert, having a low vapor pressure in their melted condition, and having good thermal density [35]. These PCM do have significant drawbacks, though. They undergo large volume change [36] and possess poor thermal conductivity that ranges from 0.1 W/mK [37] to 0.35 W/mK [38]. They also pose safety and environmental problems that limit their usage in large-scale construction applications since they are combustible and made from non-renewable petroleum resources.

Despite these obstacles, researchers explored a number of strategies to enhance the characteristics of PCM made of paraffin. To deal with the problem of high-volumetric changes, encapsulation became the most favored technique. The form and size of the paraffin droplets can be regulated by encasing them in a coating substance, which also prevents leakage and evaporation [39]. To improve heat conductivity, matrix architectures or metallic fillers have also been investigated; however, this might drive up the price of the PCM. To reduce their flammability, additional fire-retardant chemicals have also been experimented [40].

In comparison to paraffins, non-paraffin PCM which are mostly biobased PCM have better freezing and melting properties [36]. Fatty acids, esters, and glycols are examples of biobased PCM. Fatty acids having freezing temperature ranges of 17 to 64 °C and melting

temperature ranges of 16 to 65 °C are often used. Examples of fatty acids mostly studied in the literature include capric, lauric, caprylic, myristic, stearic, and palmitic acids. Biobased PCM are safe for the environment and have a number of useful qualities, including high latent heats of fusion, little supercooling, robust thermal cycling, reasonable cost, and little volume change during phase transition [12]. In addition to being reasonably priced and offering little volume change during phase transition, biobased PCM offer a workable alternative for paraffin in many TES applications [41,42].

2.2. Inorganic PCM

Inorganic PCM possess unique thermal properties which make them suitable for TES applications. Some examples of inorganic PCM include hydrated salts, oxides, and metals. Hydrated salts are among the most commonly studied inorganic PCM. For instance, magnesium chloride hexahydrate ($\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$) has been investigated for its thermal properties and stability [43–45]. Similarly, sodium nitrate (NaNO_3) has been used as a PCM for solar thermal applications due to its high latent heat of fusion and thermal stability [46,47]. Other salts that have been studied as PCM include potassium nitrate (KNO_3), calcium chloride (CaCl_2), and lithium nitrate (LiNO_3) [48–52]. Due to their affordability, comfortable phase transition temperature, and high energy storage density, inorganic PCM, in particular hydrated salts, are a family of PCM that holds out a lot of promise [53,54].

Salt hydrates have been shown to have another main advantage over paraffins which is the ease of manufacturing/disposal and lower environmental impact [55]. In spite of their numerous advantages, salt hydrate PCM have some drawbacks. For example, they may exhibit corrosive properties towards most metals and can undergo supercooling. Additionally, phase segregation during transition and phase decomposition upon numerous thermal cycling may occur, which can affect their long-term chemical stability [35,56]. To address the issues of supercooling and phase segregation in salt hydrate PCM, nucleating and thickening agents are often added [57,58]. While these agents can be effective in the short-term, there is a risk of phase separation occurring after multiple thermal cycles, which can ultimately compromise the

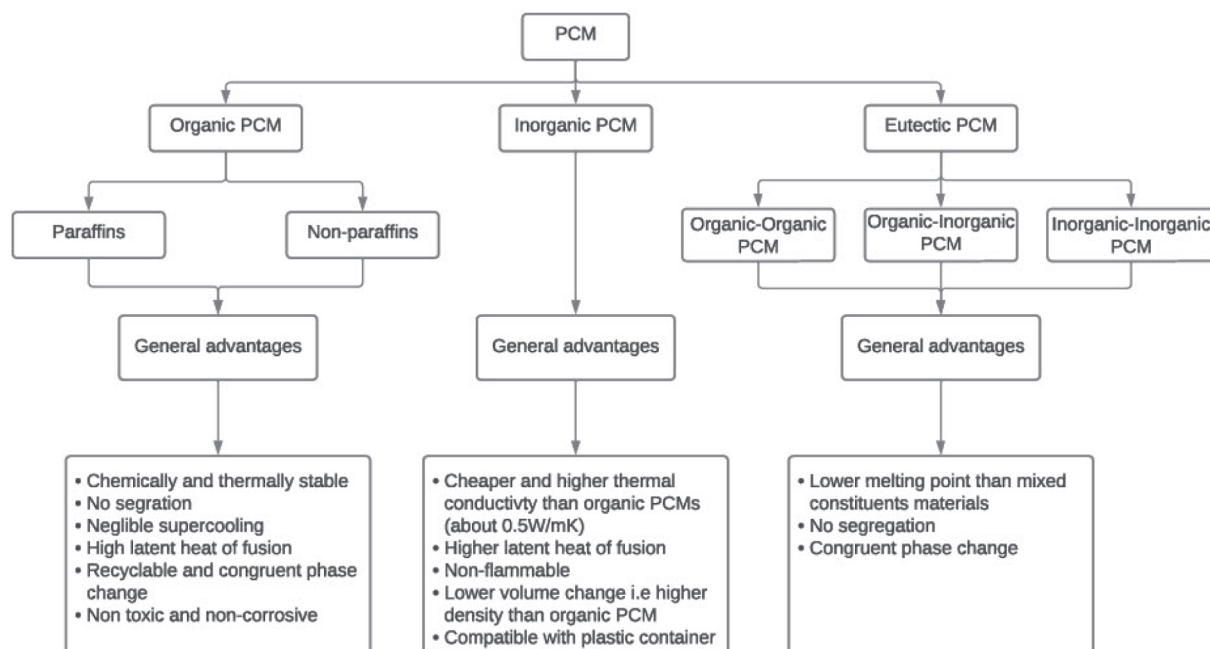


Fig. 1. Types of phase change materials and their general advantages.

performance and reliability of the PCM [59]. As such, further research is needed to develop more effective and long-lasting solutions to address these issues. Innovative techniques such as microencapsulation have emerged as potential solutions to address the issues of supercooling, phase segregation, and high-volume change associated with salt hydrates [60,61]. Microencapsulation has been shown to enhance the surface-area-to-volume ratio, thus reducing these drawbacks. Additionally, the addition of thermal conductivity agents such as expanded graphite and copper has been reported to improve the thermal conductivity of salt hydrates [62,63].

2.3. Eutectic PCM

The development of eutectic PCM has drawn a lot of interests recently as a solution to some of the major drawbacks of both organic and inorganic PCM. This involves mixing binary or ternary or more solids in proportions that gives a phase change material with improved performance than the individual constituents. Being able to store more energy with a smaller volume of material because of its special quality makes it a desirable choice for building applications. Based on the constituent materials, the eutectic PCM group can be further subdivided into three categories: organic-organic, inorganic-inorganic, and organic-inorganic. When selecting the best eutectic PCM for a particular application, it is important to consider the distinct benefits and limits that each of these groups offers. This novel approach enables the fine-tuning of crucial parameters such as transition point, supercooling reduction, and phase transition point lowering, as demonstrated by numerous studies [59]. Moreover, eutectic mixtures have shown remarkable resistance to phase separation in several instances [64], further emphasizing their potential for widespread applications. They are therefore one of the desirable alternatives for energy storage applications because of their properties, particularly where precise temperature control and frequent thermal cycling are required [4].

Table 3 shows some recent studies that have investigated the use of each of these groups of PCM and the effect of encapsulation on their performance. It can be seen from Table 3 that generally, while encapsulation slightly affects phase transition temperature, enthalpy of PCM greatly decreased after encapsulation due to the low melting point of the microcapsules and/or small amount of core material within these shells [12,65]. For the inorganic PCM, microencapsulation significantly reduced supercooling by providing nucleation sites within the shell layer [66]. While eutectic PCM provided double endothermic peaks in some studies attributable to confinement pressure from encapsulation [67] or heterogeneous mix of the PCM [68], single peak was reported in another [69]. More experimental study is needed to optimize design application needs for single or tunable peak in regions with wide temperature profiles.

3. Microcapsules

The material used for the microcapsule, also known as shells, significantly impacts the performance of the PCM. The selection of shell materials for microcapsules depends on the specific application and desired properties [74]. Common materials used for shells include organic polymers, inorganic materials, and composite (hybrid) materials. Polymers such as melamine-formaldehyde, urea-formaldehyde, and polyurethane are frequently employed as shell materials due to their good chemical stability. Inorganic materials such as silica and calcium carbonates are also used due to their high thermal conductivity and chemical stability. Composite materials that combine the advantageous properties of both polymers and inorganic materials are another option for shell materials [75,76]. However, there are certain disadvantages associated with the use of microcapsules to encapsulate PCM. Most shell materials have low thermal conductivity, which can limit heat transfer effectiveness. Additionally, the total heat storage capacity is largely reduced due to the limited mass fraction of PCM that can be stored in the

Table 3

Recent investigations involving diverse PCM, their TES benefits, and performance after encapsulation.

PCM	PCM type	T _m (°C)	Melting enthalpy of PCM (J/g)	PCM thermal properties after microencapsulation	Ref
organic	OP70 technical grade paraffin	65.2	230.8	•The low melting temperature of the shell reduced the latent heat and phase temperature of the PCM to 123.6 J/g and 57.8 °C respectively •thermal conductivity of the PCM increased linearly from 0.2 to 2.2 W/mK with the increase in expanded graphite and shell material	[65]
organic	n-octadecane	31.6	219.6	• the melting temperature of the PCM was slightly affected by the encapsulation but the enthalpy decreased up to 111.5 J/g	[70]
organic	PureTemp29	29	202	•melting temperature of the PCM was unaffected by encapsulation but caused latent heat reduction to 59.33 J/g •thermal conductivity of the MPCM was increased by the silica coating on the microcapsule	[12]
organic	RT44	52.7	257.9	•high melting point of the CaCO₃ shell delayed the phase change of the PCM by up to 6 °C and improved the thermal conductivity to about 0.85 W/mK •the enthalpy of the PCM decreased by about 31% after encapsulation	[71]
inorganic	Na ₂ SO ₄ ·10H ₂ O	37.4	211.9	• the shell did not affect the phase transition temperature of the PCM, and phase segregation of the salt hydrates was largely inhibited •enthalpy of the PCM was reduced to 125.6 J/g due to the confinement provided by the microencapsulation	[72]
inorganic	Na ₂ HPO ₄ ·12H ₂ O	43.7	251.5	• TES performance of the salt hydrate	[66]

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Table 3 (continued)

PCM	PCM type	T _m (°C)	Melting enthalpy of PCM (J/g)	PCM thermal properties after microencapsulation	Ref
				was enhanced by the encapsulation • 177 J/g was the highest enthalpy of the PCM observed after encapsulation • the supercooling of the PCM reduced due to the shell layer providing nucleation sites and the latent heat reduced by about 36% after encapsulation • the thermal conductivity of the PCM was reduced due to the low thermal conductivity of the shell material	
inorganic	Na ₂ HPO ₄ ·12H ₂ O	39.4	245.3	• the eutectic still maintained a single peak during melting after encapsulation • low thermal conductivity of the shell reduced the thermal performance of the eutectic PCM	[73]
eutectic	Na ₂ SO ₄ ·10H ₂ O- Na ₂ HPO ₄ ·12H ₂ O	29.3	218.58	• melting enthalpy of the eutectic hydrated salt reduced to 129.7 J/g • two endothermic peaks were observed attributable to the pressure of the confinement provided by the shell	[69]
eutectic	Na ₂ CO ₃ ·10H ₂ O and Na ₂ HPO ₄ ·12H ₂ O	27.3	220.2	• the PCM showed bimodal peaks after encapsulation attributed to possible heterogenous mix of the PCM rather than creating a new type of PCM • encapsulation enhanced the thermal conductivity of the PCM	[67]
eutectic	RT 28/RT 42				[68]

microcapsule [76,77].

Recent studies have focused on exploring various shell materials for MPCM, as highlighted in Table 4. The findings from these studies demonstrate that the shells provide protection and enhance thermal stability of the PCM [78]. Furthermore, the choice of shell material can impact the thermal conductivity, mechanical strength, and chemical stability of the microcapsules. A common limitation observed in shells made from organic material is the degradation of the polymer shells at temperatures above 200 °C [79,80]. Considering the flammability of this material, this may prevent their large-scale use in buildings. Similarly,

Table 4

Some recent studies on MPCM, main findings, and limitations.

Material type	Shell	Main findings	Major limitations	Ref.
organic	poly (methyl methacrylate co methacrylic acid)	- microencapsulation increased the thermal stability of the PCM by $\sim 30 \pm 2$ °C. - the crosslinker and core/shell ratio content influence MPCM formation and the enthalpy of the MPCM increased with the crosslinker percentage	- some aggregates from the polymer shell were observed between the MPCM - the shell material totally decomposed above 400 °C	[79]
organic	polyurethane (PU) made from soy polyols	- microencapsulation enhanced the thermal reliability of the PCM even after 50 thermal cycles - the microcapsule slightly delayed the decomposition of the PCM	at high core/shell ratio, the shell decomposes more rapidly	[78]
organic	polyurea, polyurethane, and polyamine	- the shell material delayed and reduced the mass loss of the MPCM by less than 6% after 1 h heating at 120 °C. - the composite shell increased the enthalpy of the MPCM	- the multiple shell caused double exothermic peaks during cooling cycle - the composite shell caused an increase in degree of supercooling of the MPCM	[81]
organic	polyurethane and polymethyl methacrylate	the hybrid binary shell stabilized the MPCM even after 50 heating/cooling cycles	- the produced MPCM are deformable under volume change during PCM phase transition - the low thermal conductivity of the shells delayed the phase transition temperature by 19.5 °C - the shell decomposed totally before 400 °C	[80]
organic	polyurea	- thermal stability of PCM was enhanced by the microcapsule. - MPCM improved the TES and thermal comfort of the chamber with PCM coatings	reduction in thermal stability of coatings with MPCM	[83]
inorganic	cenospheres	- leakage of the PCM was prevented by the shell - the silica coating on cenosphere improved the thermal conductivity of the mortar - no thermal decomposition of the microcapsule and cenosphere delayed	the thick silica coating on the cenosphere reduced the latent heat of the PCM	[12]

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Table 4 (continued)

Material type	Shell	Main findings	Major limitations	Ref.
inorganic	micro CaCO ₃	PCM decomposition by 16 °C • the CaCO ₃ shell enhanced the thermal conductivity, thermal and chemical stability of the PCM	• agglomeration of MPCM which can affect even distribution • shell degradation sets in after 240 °C	[13]
inorganic	TiO ₂	• shell material delayed thermal degradation of PCM	• agglomeration of MPCM which can affect even distribution • thermal degradation of the shell gel	[84]
inorganic	silica	• delayed thermal degradation of the hydrated salt PCM due to shell material	• agglomeration of MPCM which can affect even distribution	[66]
inorganic	GaOOH	• the microcapsule prevented the leakage of the PCM after 200 cycles and no shape deformation after 300 cycles • the latent heat of the PCM was not significantly reduced after microencapsulation	• the MPCM formation is pH sensitive • the MPCM exhibited higher supercooling than core material	[85]

cases where hybrid polymer shells are used, multiple endothermic peaks during crystallization were observed [81].

In most of the inorganic shells, the MPCM agglomerate [13,82]. This characteristic can prevent even distribution of the MPCM in construction materials. Such agglomeration was not observed in hollow micron size sphere like cenosphere [12].

3.1. Organic microcapsules

In the context of building materials, the utilization of organic materials for MPCM has been a subject of significant research, as highlighted in Table 5. It can be seen from Table 5 that majority of commercially available MPCM have polymer shells. This preference can

be attributed to the structural flexibility of polymer shells, which impart inertness to the PCM [86]. However, it is important to recognize that organic have low mechanical stiffness [87,88], high flammability, and low thermal conductivity [84,89]. The low mechanical property of polymers is a huge concern for applicability in construction materials and must be thoroughly investigated for the integrity of the intrinsic mechanical performance of cementitious composites. One approach to improve the polymer shell performance is the incorporation of various fillers into the polymer shell, thereby augmenting their mechanical and thermal attributes [90].

3.2. Inorganic microcapsules

Inorganic microcapsules exhibit higher mechanical strength, improved thermal conductivity, and increased stiffness compared to their organic counterparts [38]. These enhanced characteristics contribute to the development of MPCM with exceptional thermo-mechanical properties. Compendium of some recent studies with inorganic shell and their thermophysical properties are shown in Table 6. It

Table 6

Summary of common inorganic microcapsules for MPCM studied in the last decade.

Microcapsules	PCM	Microcapsule size (μm)	T _m (°C)	Latent heat (kJ/kg)	Ref.
ZnO	n-eicosane	4–8	38	–	[101]
Cenospheres	PureTemp 29	10–200	29.9	59.3	[12]
TiO ₂	n-octadecane	2–5	25.6	31	[84]
Silica shell	Na ₂ SO ₄ ·10H ₂ O	0.5–28	33.6	125.6	[72]
Silica shell	Na ₂ HPO ₄ ·12H ₂ O	–	57.4	177	[66]
CaCO ₃	RT 28/RT 42	1–5	19.8/ 34.8	82.8	[68]
Silica shell	Stearyl alcohol	5–12	55.9	112.1	[107]
TiO ₂	Myristic acid	–	52.2	97.7	[108]

Table 5

Summary of common organic microcapsules for MPCM studied in the last decade.

Microcapsules	PCM	Microcapsule size (μm)	Melting temperature (T _m , °C)	Latent heat (kJ/kg)	Ref.
Polymethylmethacrylate (PMMA)	Micronal DS	5	26	110	[91,92]
Acrylic polymer	Micronal 5040X	200–800	23	100	[93]
Acrylic polymer	Micronal 5038X	200–800	26	110	[93,94]
Polymer shell	Micronal DS 5001 X	100–300	23	110	[95]
Polymer shell	Paraffin	17–20	37	–	[96]
PMMA	Paraffin	50–300	23	100	[97]
Copolymer	Paraffin	5	24	70	[98]
Melamine-formaldehyde	Paraffin	15–20	24	160	[99]
Polyurethane and polyurea	Paraffin MPCM	1–20	41.4	192	[100]

can be seen that this shell is suitable for any PCM and size of the microcapsules are relatively smaller than organic shells.

In the existing literature, two common methods are employed to produce inorganic shells. The first method involves developing the microcapsule around the core material, as demonstrated in ZnO [101], TiO₂ [84,102], SiO₂ [66,72], and CaCO₃ [68,71]. This approach typically involves a chemical process supplemented with mechanical stirring and pH adjustments to form the MPCM. The second method involves impregnating the PCM into pre-existing hollow micron-sized shells, such as cenospheres [12]. Silica (SiO₂) shells are widely used for MPCM incorporated in cementitious composites. This is due to the potential bonding between silica microcapsules and calcium hydrate during cement hydration, forming additional calcium silicate hydrate (C-S-H) gel [103]. Additionally, silica enhances the thermal conductivity of MPCM [104].

Drawback of the first method is that the shell material may separate from the PCM during thermal cycling [105]. On the other hand, the second approach, produce shells with better bonding with PCM, and no damage during thermal cycling. This is attributed to the thick walls of the shell material, typically a few micrometers thick, such as cenospheres with a crushing strength of approximately 10–20 MPa [106]. However, the surface of these hollow micron-sized shells must be coated or sealed to prevent leakage of melted PCM through surface perforations [12]. Inorganic shells compensate for the significant reduction in mechanical properties often observed in construction materials incorporated with MPCM. This advancement holds great potential for widespread application of MPCM in buildings, enabling improved thermal management and a reduction in operational carbon emissions.

3.3. Hybrid microcapsules

In recent years, researchers have been exploring the potential of hybrid microcapsules, which combine organic and inorganic materials, to overcome some major limitations associated with each material type [82]. Zhao et al. [82] focused on a bifunctional hybrid shell composed of n-octadecane PCM encapsulated in TiO₂-doped PMMA. The addition of TiO₂ enhanced the thermal conductivity of the PMMA shell, reduced supercooling by 1.33 °C at 10% TiO₂, and increased the onset thermal decomposition temperature of the shell by approximately 17 °C at 6% TiO₂. However, energy efficiency of the MPCM decreased with increasing amount of TiO₂. Wang et al. [109] investigated a PMMA-silica hybrid shell with n-octadecane as the core material. The researchers found that increasing the amount of monomer improved the mechanical stability of the microcapsule. The produced microcapsules exhibited favorable thermal performance during storage-release cycles, as well as enhanced thermal stability and reliability.

Table 7
Summary of common hybrid microcapsules for MPCM studied in the last decade.

Microcapsules	PCM	Microcapsule size (μm)	T _m (°C)	Latent heat (kJ/kg)	Ref.
PMMA-SiO ₂	n-octadecane	5–15	26.3	122.3	[109]
PMF-SiC	n-octadecane	3	29.5	168.0	[105]
PMMA/TiO ₂	n-octadecane	10–20	21.1	106.6	[82]
PMMA/TiO ₂	n-octadecane	3–16	30.9	179.9	[110]
PMMA/polyurea	n-hexadecane	17–20	–	210.7	[111]
Polymer/SiO ₂	Paraffin	4–16	27	184.4	[112]

Summary of other recent studies in the literature as shown in Table 7. One very important observation from Table 7 is that hybrid shells are often used to encapsulate organic PCMs. This may be because the production method involves oil–water emulsion and heating which can contaminate/affect the crystallization water in inorganic hydrated salts. Due to the cost effectiveness of inorganic PCM, further research needs to be carried out on how to encapsulate inorganic PCM within hybrid shells. This can create economical MPCM that may be widely accepted and address some of the problems of salt hydrate PCM.

4. Manufacturing process of MPCM

The literature offers a plethora of techniques for producing MPCM, with the specific method chosen depending on the desired application. However, the ultimate objective remains consistent: to prevent the leakage of melted PCM and enhance the thermal and mechanical performance of the MPCM. The three main production techniques of MPCM are: physico-mechanical, chemical, and physio-chemical methods, as depicted in Fig. 2. This review will specifically focus on the frequently utilized microencapsulation techniques for MPCM used for thermal management of buildings.

4.1. Physico-mechanical method

This method, although not commonly used for MPCM in building applications, is prevalent in the pharmaceutical and food industry. It involves the mechanical production of the shell material without undergoing any chemical reaction, relying solely on physical processes such as dehydration, adhesion, and drying. Among the techniques used, spray drying is the most extensively utilized for creating small round microcapsules with a desirable particle size.

4.1.1. Spray drying

The process of spray drying involves a typical setup as shown in Fig. 3 [113]. A homogenized emulsion containing the core material, coating materials, and solvent is prepared and fed into the atomizer. Compressed N₂ is used to atomize the emulsion at a controlled flow rate. The atomization process is followed by solvent evaporation in the drying chamber, resulting in the formation of the microcapsules, which are then collected at the end. In a study by Borreguero et al. [113], paraffin wax RT27 with and without carbon nanofibers was encapsulated in a polyethylene-ethylvinylacetate polymer shell using spray drying. They observed that the developed microcapsules were homogeneous, and the addition of carbon nanofibers enhanced the thermal conductivity of the shell. However, a major limitation they encountered was the agglomeration of some microcapsules in the drying section, which can be mitigated by using a high stream gas carrier at the atomizer or reducing the drying chamber temperature. Another study focused on encapsulating n-octadecane and methyl palmitate in a silica shell using the spray drying process [114]. The researchers found that the concentration of surfactant influenced the emulsion droplets and encapsulation efficiency during atomization.

In this method, determining the optimum amount of PCM is crucial to ensure effective microencapsulation and prevention of fragile and collapsible microcapsules. While the spray drying technique offers scalability and cost-effectiveness, the equipment itself can be expensive, potentially increasing the overall cost of producing MPCM. It is important to adjust the temperature of the drying chamber below the transition temperature of prevent melting of the PCM. Generally, this method is more suitable for producing microcapsules that are not subjected to large and repeated thermal cycles due to their fragility and non-uniformity. This could explain why it is not commonly used for manufacturing MPCM for thermal energy storage in building materials.

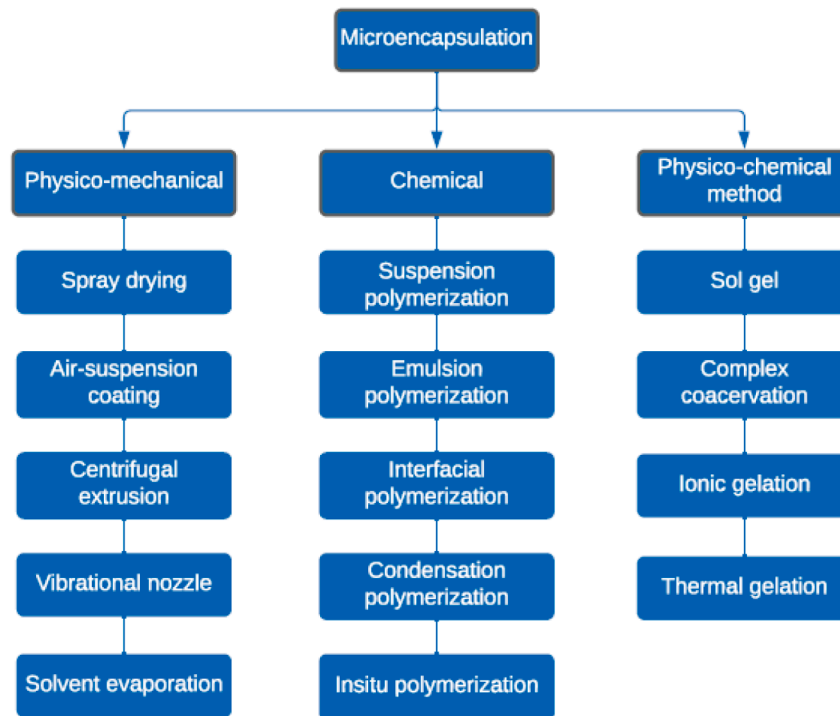


Fig. 2. Microencapsulation techniques.

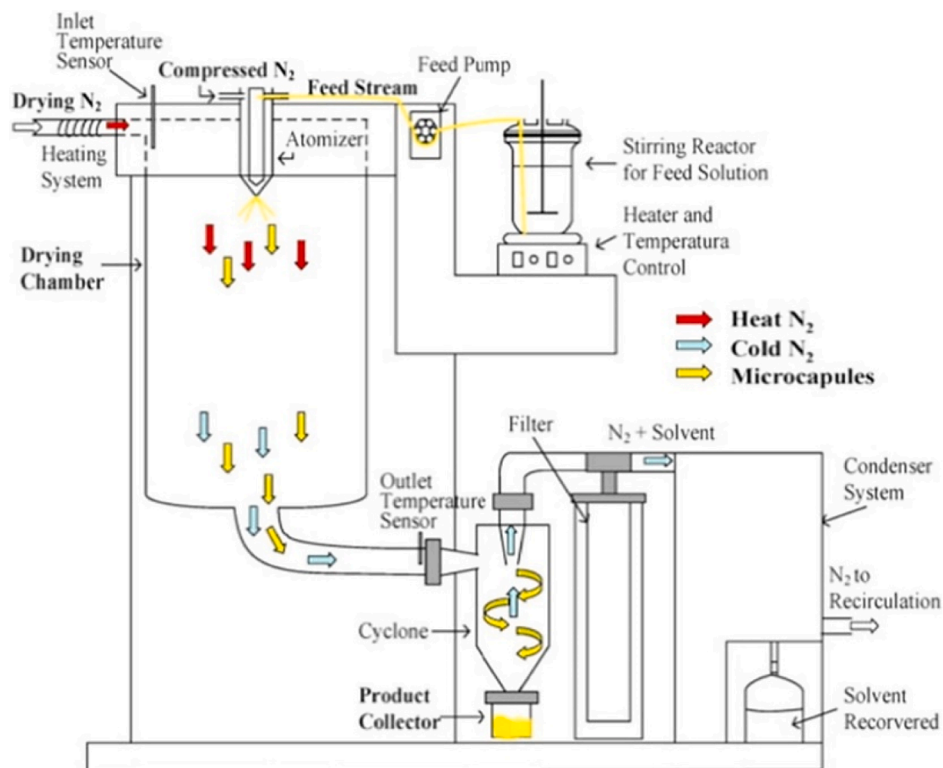


Fig. 3. Schematic representation of the spray drying equipment [113].

4.2. Chemical method

The chemical method is the most used method for the fabrication of

MPCM used in the cementitious material. This can be because it offers a better technological approach in terms of encapsulation efficiency and structural integrity of the core material [76]. In this method, the shell is

formed around the core material through chemical processes such as suspension polymerization, emulsion polymerization, in-situ polymerization, condensation polymerization, and interfacial polymerization. The microcapsule material can be organic, inorganic, binary or ternary organic or inorganic shells, or hybridization of organic–inorganic materials. This method often times are carried out under temperature and pH controlled experimental set up. Generally, the MPCM are formed through the polymerization and condensation process of monomers to form shell around a core material. The widely adopted chemical methods for MPCM production are further discussed.

4.2.1. Suspension polymerization

This method involves a heterogeneous radical polymerization procedure with mechanical agitation. The dispersed phase consists of the core material, monomers, and a monomer-soluble initiator to initiate the polymerization reaction. Initially, the dispersed phase is mixed with the continuous phase, typically water, under mechanical agitation to form a suspension. The polymerization reaction takes place in the small polymerizable droplets, resulting in the formation of microcapsules. It is important to note that for successful MPCM production using this approach, the monomer should be insoluble or fairly insoluble in water to ensure proper dispersion. The radical initiator used to initiate the reaction process is typically activated by adjusting the pH or temperature of the medium [115].

In a study by Li et al. [116], the suspension polymerization method was employed to produce copolymer microcapsules containing various amounts of n-octadecane. The researchers investigated the impact of initiator type and polymerization temperature on the resulting MPCM. This approach successfully produced spherical microcapsules. They observed that the water-soluble initiator, ammonium persulfate, provided better shells compared to the oil-soluble initiator, 2,2 azobisisobutyronitrile. Another recent study focused on the development of styrene–divinylbenzene copolymer shells around n-octadecane using suspension polymerization [117]. The TES potential of the resulting MPCM was examined at different weight percentages in cementitious building boards. The prepared MPCM exhibited spherical shape with smooth and complete encapsulation of the PCM. The research findings indicated that as the shell-to-core ratio increased, the degradation temperature of the MPCM also increased. This was attributed to the thicker shell providing greater resistance to the decomposition of the PCM. In another study, styrene–divinylbenzene copolymer doped with titanium dioxide nanoparticles to encapsulate n-octadecane using suspension polymerization [118]. The results showed successful encapsulation of the n-octadecane, with spherical microcapsules exhibiting a rough surface, good dispersion, an encapsulation rate of $48.0 \pm 0.4\%$. Furthermore, microcapsules containing 4.0 wt% TiO_2 exhibited the highest thermal storage capacity, with molten and crystallized enthalpies of 105.3 ± 0.3 and 102.4 ± 0.7 J/g, respectively. The optimum amount of MPCM to be added into the cementitious material should be determined. Zhao et al. [117] demonstrated that with 30 wt% MPCM, the TES of a cement board increased by 67.82% compared to the control. However, at incorporation levels exceeding 30 wt% of MPCM, the board became weak and prone to breakage during demolding.

4.2.2. Emulsion polymerization

Emulsion polymerization is another form of radical polymerization that does not require mechanical agitation. It differs from suspension polymerization in that it often utilizes a water-soluble initiator instead of a monomer-soluble initiator. The typical steps involved in emulsion polymerization are as follows: (i) the monomers PCM are emulsified in a dispersion medium using surfactants, and (ii) an aqueous solution of a radical initiator is added to initiate the polymerization reaction. This method is commonly used for polymer microcapsules.

Enteshari et al. [119] synthesized melamine formaldehyde microcapsules through emulsion polymerization and investigated the thermal resistance of the developed shell. They found that the encapsulation

efficiency of the shell was 73.5%, and the microcapsules exhibited thermal stability even after repeated thermal cycles. In a study by Mahajan et al. [79], the emulsion polymerization technique was employed to create PMMA microcapsules with n-tetradecane core. They varied the quantity of crosslinking agents from 3 to 20% w/w, the shell-to-core ratio from 1:1 to 1:2, and the initiator percentage from 1 to 3. The results showed that the MPCM had a higher enthalpy of around 127.3 J/g and 66.72% efficiency at 20% w/w crosslinking amount and the thermal stability was approximately up by 30 °C. In another study, palmitic acid encapsulated in a poly (allyl methacrylate) shell was prepared via emulsion polymerization and tested for thermal performance in mortar [120]. The synthesized MPCM exhibited a good latent heat capacity ranging from 143 to 188 J/g, and the mortar containing the MPCM showed a temperature 6.6 °C lower than the control mortar. Su et al. [121] also utilized the emulsion polymerization technique to prepare graphene-based MPCM. The thermal conductivity of the MPCM reached 7.2 W/(m·K) due to the presence of graphene. The addition of graphene significantly reduced supercooling and decreased the leakage rate by > 90%.

Similar to the suspension polymerization method, the emulsion polymerization technique is effective in producing spherical microcapsules that encapsulate PCM. The choice of surfactant used can influence the morphology of the resulting microcapsules.

4.2.3. In-situ polymerization

In this method, the PCM is added to an aqueous solution containing surfactant and agitated to form a homogeneous oil/water (O/W) emulsion. A shell material prepolymer solution is then prepared from monomer pre-polymerization and added to the O/W emulsion under controlled temperature and pH to synthesize the microcapsules.

Liu et al. [122] utilized the in-situ polymerization method to synthesize phenol–formaldehyde resin shells. They investigated the influence of emulsification concentration, core/shell ratio, and agitation speed on the morphology and thermal properties of the MPCM. The microcapsules produced exhibited a relatively spherical shape with a well-defined core–shell structure. A core/shell ratio of 3:6.47, at an agitation speed of 1000 rpm and emulsification concentration of 1%, resulted in desirable microcapsule characteristics. The microcapsules also demonstrated thermal reliability even after 500 thermal cycles. In another study, the thermophysical properties of stearyl alcohol encapsulated in melamine formaldehyde shells produced through in-situ polymerization were investigated [123]. The results revealed that the surfactant content in the emulsion influenced the morphological structure of the microcapsules. The developed MPCM exhibited good thermal performance and chemical stability without any leakage of the melted PCM. Furthermore, Park et al. [124] studied the effect of MPCM on indoor thermal comfort and energy usage. The microcapsules, embedded in wallboards, were prepared using the in-situ polymerization method and successfully encapsulated approximately 70% of n-octadecane. The inclusion of MPCM in the wallboard led to a reduction of 1–2 °C in room temperature fluctuations, energy consumption of the system decreased by 27.7%. In the production of magnetized graphene oxide-modified MPCM, this microencapsulation technique was also employed to enhance heat transfer and reduce melted PCM leakage [125]. The produced MPCM exhibited a spherical shape with an average diameter of 5 μm . The encapsulation ratio reached 88.28%, and the leakage rate was reduced by 60% compared to unmodified MPCM.

Overall, the microcapsule morphology and properties using this method can be influenced by factors such as emulsification concentration, core/shell ratio, and agitation speed.

4.2.4. Interfacial polymerization

Interfacial polymerization stands out as a method that occurs at the interface of two immiscible phases, offering unique advantages and applications. The process typically involves the addition of PCM and shell material to an aqueous solution of surfactant, creating an oil-in-

water emulsion. Hydrophilic monomers are then introduced for polymerization at the core-shell interface, resulting in the formation of microcapsules.

Zhang et al. [126] found that the pH of the reacting solution played a significant role in the formation of the microcapsules using this technique. At a pH of 2.89, they observed the development of compacted silica shells with a relatively large mean particle size of approximately 17 μm . These silica shells greatly enhanced the thermal conductivity of the core material. Similarly, Yan et al. [127] employed the interfacial polymerization technique to encapsulate eicosane within a polyurea shell. The resulting MPCM exhibited a regular spherical shape, showed high enthalpy of 209.8 J/g and an encapsulation ratio of 82.9% making them promising candidates for thermal management applications. Further advancements were made by Lu et al. [81], who utilized the interfacial polymerization technique to prepare three different types of MPCM with polyurea, polyurethane, and polyamine shells. The microcapsules generated through this method displayed spherical shapes with intact and compacted surfaces, showcasing high thermal stability. The stirring velocity during the emulsification stage proved to have a significant impact on the size of the microcapsules, with higher stirring speeds resulting in smaller particles and more uniform distributions. The core-to-shell ratio also played a crucial role, with ratios below 4 yielding high core content and encapsulation efficiency. The inclusion of a three-composition shell further enhanced the density of the microcapsules, ensuring less than 6% mass loss after drying. The potential applications of MPCM prepared through interfacial polymerization was explored in the field of multifunctional pumping wet shotcrete (PWS) by Lu et al. [128]. By encapsulating paraffin wax with a poly(lauryl methacrylate) shell, they demonstrated outstanding performance and effective utilization in PWS. The synthesized MPCM not only reduced the thermal conductivity of PWS but also provided the ability to regulate heat, highlighting their versatility and practical value.

4.3. Physico-chemical method

Within the realm of microencapsulation techniques by physico-chemical method, two commonly employed methods are the sol-gel process and complex coacervation. These approaches blend physical processes, such as heating, cooling, and phase separation, with chemical processes like condensation and cross-linking to create MPCM. While complex coacervation is typically utilized for organic shells, the sol-gel technique predominantly serves as a means for developing inorganic shells. Complex coacervation offers precise control over microcapsule size and thickness, but it does have its limitations. The shells produced through this method can sometimes agglomerate, and scaling up the process proves challenging [88]. As a result, this section will focus solely on the sol-gel method, which offers its own unique advantages and applications.

4.3.1. Sol-gel method

The sol-gel method is a widely employed technique for synthesizing inorganic shells, particularly silica, due to its simplicity and efficiency. The process involves several steps: (i) hydrolysis of the precursor (usually metal alkoxides) in an aqueous solution, acid, or base, (ii) addition of the phase change material (PCM) into a surfactant-based aqueous solution, resulting in the formation of micelle solution, (iii) combining the hydrolyzed precursor solution with the micelle solution and initiating polycondensation to form microcapsules.

Zhang et al. [129] utilized the sol-gel method with tetraethyl orthosilicate as the precursor and n-octadecane as the PCM to produce silica microcapsules. The resulting shells, formed at a pH of 2.45, exhibited smooth and compact surfaces, effectively enhancing the thermal conductivity of the encapsulated n-octadecane PCM. Similarly, a sol-gel approach was employed to encapsulate Sn metallic PCM within silica shells [130]. The resulting microcapsules possessed spherical shapes with a wall thickness of approximately 2 μm and demonstrated

excellent thermal performance even after 100 cycles. Furthermore, a study investigated the use of sodium silicate as a precursor for synthesizing silica shell microcapsules to encapsulate paraffin via the sol-gel technique [131]. Under specific conditions (pH 3 and core/shell ratio of 5:1), a well-structured spherical MPCM was successfully synthesized, exhibiting an encapsulation ratio of 74.14% and a high latent heat of approximately 186.4 J/g. The MPCM displayed exceptional thermal stability, performance, and durability. In another study, the sol-gel technique was utilized to develop a TiO_2 shell around an inorganic molten salt, KNO_3 [132]. The resulting MPCM demonstrated a high encapsulation ratio and efficiency, reaching 70.3% and 69.3%, respectively. Remarkably, the thermal performance and reliability of the MPCM remained excellent even after 100 repeated thermal cycles. These examples highlight the effectiveness of the sol-gel method in producing inorganic microcapsules with enhanced thermal properties and durability. Summary of these encapsulation techniques, their advantages and disadvantages are as shown in Table 8 below.

5. Enhancing the thermal properties of PCM and shell material

To fully harness the potential of MPCM for thermal energy storage in practical applications, addressing the significant limitation of low thermal conductivity of either the PCM or the shell or both is crucial. In the literature, researchers have explored different ways and proposed innovative methods to enhance the thermal conductivity of MPCM. These methods involve (1) incorporating conductive nanoparticles into the PCM, (2) introducing conductive nanoparticles into the microcapsule shell or modifying the surface of the microcapsules.

A comprehensive review of studies conducted in the past decade is presented in Table 9, showcasing promising advancements in the thermal properties of MPCM. From Table 9, it is evident that carbon nanotubes (CNTs) are frequently utilized and have undergone testing for inorganic PCM [133], organic PCM [134], and shell material [135] thermal improvement. This is because CNTs have excellent thermal conductivity and tensile strength and their nanoscale particle size make them easily blend with the PCM or shell material. The main drawback of adding these thermal enhancement nanoparticles into PCM is the reduction in latent heat of the PCM at higher wt.% incorporation level [136,137]. Therefore, it is crucial to determine the optimum amount of these particles that will offer the best PCM thermal performance. Another critical observation in Table 9 is that most of the shell materials being doped with these thermal enhancers are organic. This can be attributed to the ease of integrating these nanomaterials into the emulsion during the production process of organic microcapsules.

It is important to note that the incorporation of nanomaterials and the implementation of modified microcapsule production processes may lead to increased costs associated with MPCM. Inorganic PCM, considering its lower cost advantage over organic PCM may be a possible economical solution. Nevertheless, the pursuit of these advancements is essential for realizing the widespread utilization of MPCM in various applications.

6. MPCM incorporation into building material

Materials must meet up with requirements established by design regulations and industry standards in the construction industry. These are part of the safety measures put in place to avoid premature failure of the material during its service life and prevention of collapse which could lead to colossal loss of lives and properties. Any modifications to these materials must undergo careful analysis and comparison with their known properties, behavior, and performance. The introduction of MPCM into building materials has the potential to impact and affect various intrinsic properties of the materials. In this section, we unravel the transformative effects of MPCM on the properties of cementitious materials and wallboards.

Table 8
MPCM production techniques, advantages, and disadvantages.

Technique	process	Shell material	Shell quality/thermal performance	Advantages	Disadvantages
Physico-mechanical	spray drying	organic/inorganic	poor/poor	fast, easy scalability, and cheap	Agglomeration of microcapsules, expensive equipment
chemical	suspension polymerization	organic	good/poor	smooth spherical shells	low mechanical strength shells
	emulsion polymerization	organic	good/poor	easy to control	low mechanical strength shells
	in-situ polymerization	organic/inorganic	good/good	compact shell	low mechanical strength shells
	interfacial polymerization	organic	good/good	easy to control	low mechanical strength
Physico-chemical	sol-gel method	inorganic	good/good	spherical microcapsules with high thermal conductivity	expensive chemicals are used

Table 9
Some studies on enhancement of MPCM for thermal energy storage.

PCM	Shell material	Enhancement method	Summary of findings	Ref
stearyl alcohol	urea-formaldehyde	carbon nanotubes (CNTs) / stearyl acid	At 4% CNT, the MPCM thermal conductivity increased by 79.2 percent, thermal breakdown was delayed by 38 °C, and durability and thermostability were all improved.	[134]
paraffin	melamine/urea-formaldehyde	doped shell material with Graphene/CNTs/nano-Ag respectively	The MPCM with 0.02% graphene has the fastest thermal response rate, and supercooling can be decreased to 0.5 °C at core/shell ratios of 1/2 and lower as core/shell ratios decline.	[135]
n-octadecane/n-octacosane	melamine formaldehyde /resin (MF resin)	octadecyl isocyanate- grafted CNTs added to the PCM	The enhanced PCM only experiences a tiny amount of supercooling, and the thermal conductivity of the MPCM increases by 71.4%. (ii) With 3 wt % of the enhancer, the PCM's latent heat is 187.9 J/g.	[138]
n-octadecane	PMMA	Silicon nitride / PMMA	The MPCM demonstrated improved thermal stability, dependability, and high TES, (ii) Increasing silicon nitride from 5% to 30% increased the MPCM's thermal conductivity by 58.6%.	[139]
paraffin	poly (methyl methacrylate-co-methyl acrylate)	modified the shell with nano alumina (nano-Al ₂ O ₃)	With a thermal conductivity of 0.3104 W/mK, 16 percent nano-Al ₂ O ₃ provided the best thermal performance. Additionally, the alteration increased the thermal stability.	[140]
n-hexadecane / n-octadecane separate cores	poly(urea-formaldehyde)	shell for each PCM were modified by silver nanoparticles	The thermal performance of n-hexadecane microcapsules was enhanced by nano-Ag by 121% (0.0557 to 0.1231 W/mK) and by 41% (0.0695 to 0.0978 W/mK).	[141]
paraffin	poly(melamine-formaldehyde)	CNTs were added to the PCM	0.015 wt% CNT in 30% Heat transmission was boosted by PCM-plaster composite by 60%, and PCM's thermal buffer performance was improved.	[142]
Rubitherm (RT21)	PMMA	PMMA/pretreated with poly (dopamine)/coated with silver	The thermal conductivity of the MPCM was greatly improved by the silver coating, going from 0.189 to 2.41 W/mK. Additionally, the apparent thermal conductivity of the MPCM is related to the size of the microcapsule.	[143]
RT4	–	different % of CNTs added to RT4	An increase in CNT mass fraction improved the thermal conductivity of the composite PCM, and (ii) An increase in CNT mass fraction lowered the phase transition temperature and latent heat of the PCM.	[136]
dodecanol	MF resin	Shell/graphene oxide/CNTs	Thermal conductivity rose by 195% at 0.6 wt% of the hybrid nanoparticle, with a modest decrease in latent heat.	[137]

6.1. Cementitious composites fresh and hardened properties

MPCM and their impact on the properties of fresh and hardened cementitious composites has captivated the attention of researchers worldwide. Over the years, numerous studies have meticulously explored and examined the outcomes, shedding light on the exciting possibilities that lie within. The results from these studies are discussed and evaluated in the following sections. In addition, some research areas that require further studies are highlighted.

6.1.1. Workability

For the ease of constructability and placement of cementitious materials, it is crucial that they have good workability. However, the incorporation of MPCM into mortar or concrete has been found to significantly affect their workability, often times resulting in a reduction. This has been discussed in numerous published literature studies, and several stated reasons, as shown in Fig. 4. One reason is the size of the microcapsules. The presence of small microcapsules can lead to

increased viscosity and hinder the flowability of the cementitious material. Also, the water adsorption susceptibility of the shell, especially in polymeric shells, can contribute to the reduction in workability. The polymeric shells may absorb water from the cementitious matrix, reducing the water needed for hydration consequently leading to decrease in workability. The morphology of the microcapsule surface is another factor. If the surface of the microcapsules is rough or irregular, it can interfere with the flow of the cementitious material, reducing its workability. The properties of the shell material also play a significant role. Some shell materials may have higher stiffness or viscosity, which can negatively impact the workability of the cementitious material. Furthermore, the leakage of PCM into the cementitious matrix can act as a lubricant, reducing the friction between particles and resulting in a loss of workability.

Cunha et al. [144] investigated the physical and mechanical properties of mortars based on different binders. They studied hydraulic lime, aerial lime, gypsum, and cement, each incorporating 0%, 20%, 40%, and 60% by mass of sand. They observed an increase in the water-

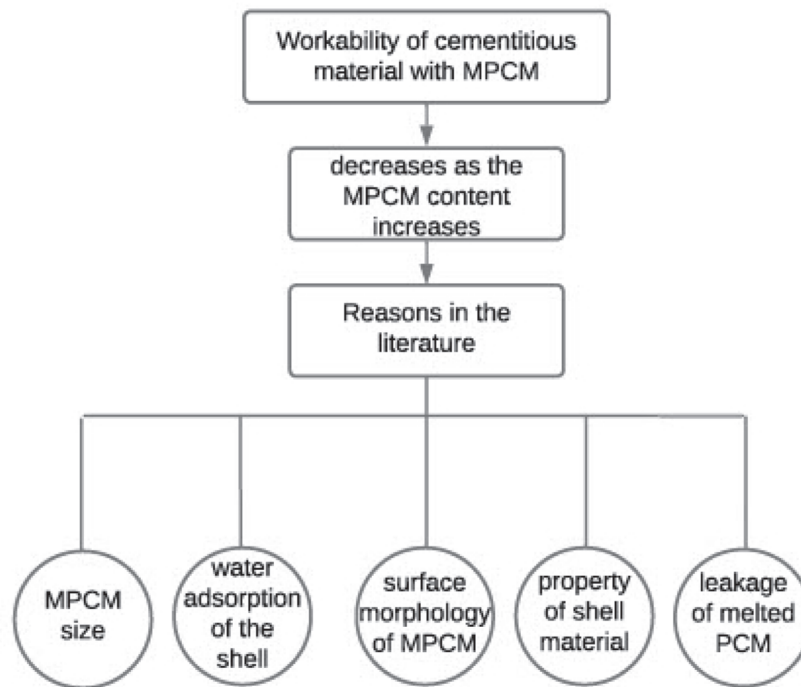


Fig. 4. Effect of MPCM on workability of cementitious material.

to-binder ratio as the MPCM level increased. At 20% MPCM incorporation, the water content increased by approximately 5%. The reduction in workability was attributed to the reduced particle size of the MPCM and the water absorption characteristics of the polymeric shell. Additionally, their findings showed that gypsum-based mortars exhibited a higher water-to-binder ratio compared to other binders, while aerial lime-based mortars demonstrated lower water content compensated for by superplasticizers. In another study evaluating the printability and thermal performance of 3D concrete with MPCM, it was found that the water-cement ratio needed to be increased as the MPCM content increased [145]. The water-cement ratio was adjusted to 0.42, 0.60, and 0.72 for 0%, 10%, and 20% MPCM, respectively, to achieve better workability and printability. The study showed that the fluidity of the mixture at 10% MPCM was higher than at 20% MPCM, which was attributed to the MPCM particles absorbing free water due to their high specific area, small size, and density.

Sanfeliix et al. [146] conducted a study to examine the effect of MPCM on the flow behavior of cement composites. They performed a mini-slump test on cement pastes containing 15%, 30%, and 45% by weight of MPCM with hydrophilic, amphiphilic, and hydrophobic surfaces. The results showed that the inclusion of MPCM led to a decrease in the mini-slump, indicating a decline in workability of the paste. This reduction in workability was attributed to the higher concentration of solid components and water adsorption on the surface of the MPCM particles. They also observed that the type of shell material influenced the workability. Hydrophobic surface MPCM exhibited workability close to that of the reference mortar at 15% incorporation level, which was higher compared to hydrophilic and amphiphilic surface MPCM. The workability of the amphiphilic MPCM was better than that of the hydrophilic microcapsules, which was attributed to agglomeration and water trapped within these agglomerated microcapsules. They concluded that small particles with a large surface area and hydrophilic surface readily absorbed water.

Leakage of melted PCM into the cementitious matrix and the

adsorption of water by the hydrogen bonds of the amino and hydroxyl groups of the shell were also identified as reasons for the reduced workability of cementitious composites with MPCM [147,148]. Alsaa-dawi et al. [149] investigated the thermal, mechanical, and micro-structural properties of normal and high-strength concrete, as well as normal and high-strength lightweight concrete incorporating 1%, 3%, 5%, 7%, and 10% MPCM. They tested the fresh properties of all the concretes using slump test and found that the incorporation of MPCM caused a reduction in workability. The slump of normal strength concrete with a water-to-cement ratio of 0.5 and 1% superplasticizer decreased from 135 mm (reference mix) to 72 mm (10% MPCM mix). Similarly, the slump of high-strength concrete with a water-to-binder ratio of 0.25 and 3% superplasticizer decreased from 172 mm (reference mix) to 106 mm (10% PCM mix). Generally, lightweight concrete exhibited slightly lower workability compared to normal weight concrete, which was attributed to the higher water absorption capacity of the lightweight aggregate affecting water content and workability.

In another study, the effect of mixing speeds on the compressive strength degradation mechanism of cement paste with MPCM was studied [150]. Cement pastes with 0% and 10% MPCM were prepared at a fixed water-cement ratio of 0.4 and mixing speeds of 70, 290, and 2000 rpm. The researchers found that moderate to high-speed mixing resulted in damage to the MPCM, which affected the workability of the mix. The height of the sample after collapse was measured to assess the workability, Fig. 5. It was observed that the cement pastes and MPCM cement pastes mixed at 70 rpm (Fig 5a and 5b) exhibited lower workability compared to pastes mixed at 290 rpm and 2000 rpm. The MPCM-70 paste appeared stiffer compared to the cement-70 paste, which was attributed to the higher solid content and water absorption capacity of the MPCM. However, when the mixing speed ≥ 290 rpm, the MPCM paste displayed a reduced collapsed height compared to the corresponding cement paste, as shown in Fig. 5b and 5e. This observation was attributed to the leakage of PCM, which acted as a lubricant in the paste.

To address the workability issue caused by the incorporation of

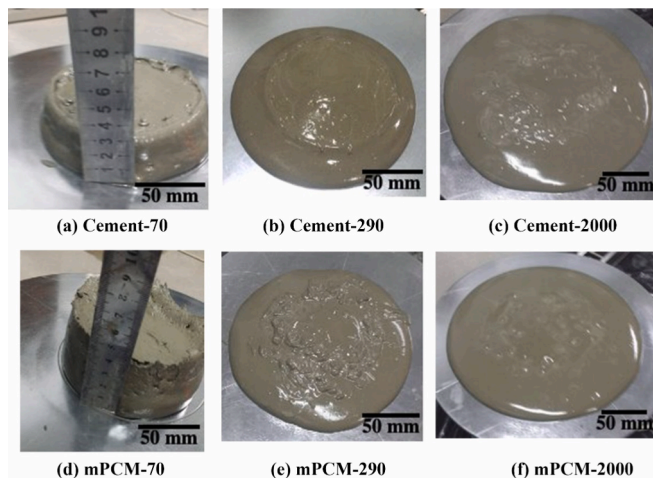


Fig. 5. Observation made on pastes before dropping the flow table [150].

MPCM, the water-cement ratio can be increased as the MPCM content increases, or superplasticizers can be added to the mix to compensate for the workability loss [87,151,152]. However, it is important to note that modifying the water-cement ratio may have a negative effect on the compressive strength of the cementitious composite. Similarly, the addition of superplasticizers can also affect the setting time of cementitious composites. Therefore, it is essential to determine the optimum amount of water or superplasticizer to be added to the mix in order to prevent any deleterious effects on the overall performance of the cementitious composite.

6.1.2. Hydration of cementitious material

The heat of hydration in cementitious materials is generated as a result of the chemical reaction that occurs during the hydration process. In large concrete structures, such as mass concrete, the heat generated can lead to high temperatures, reaching up to 60 °C. This temperature gradient between the surface and interior of the structure can potentially cause thermal stresses resulting to cracks. Therefore, it is crucial to control the heat of hydration. There are several contemporary methods to control the heat of hydration [153]: One approach is to use concrete with a low starting temperature, which can help reduce the overall temperature rise during hydration. Another method is to partially replace Portland cement with cementitious materials that have slower reaction rates, which can effectively reduce the heat generation. Additionally, the overall cement content can be reduced through the use of chemical admixtures or by combining these different approaches.

A recent approach to controlling the heat of hydration in cementitious composites is the incorporation of MPCM. The temperature rise during cement hydration can be reduced due to the endothermic reaction of the PCM during its melting phase. This ability of PCM to absorb heat can help mitigate the maximum thermal stress caused by cement hydration in massive concrete structures and result in fewer thermal cracks in the cementitious composite [154]. Furthermore, MPCM offer the advantage of early-age curing in cementitious composites [35]. To measure the hydration rate and heat of hydration, isothermal calorimetry and semi-adiabatic calorimetry are commonly used techniques [155]. These methods allow for the monitoring and quantification of the heat released during the hydration process, providing valuable insights into the hydration kinetics and the effectiveness of heat control measures.

There are conflicting reports in the literature regarding the effect of MPCM on cement hydration and the resulting heat of hydration. Some studies have shown that the inclusion of MPCM in cementitious composites successfully reduced and delayed the heat of hydration. For example, Eddahhak et al. [156] measured the heat of hydration in mortars with integrated MPCM using semi-adiabatic calorimetry. They

found that MPCM effectively reduced the heat of hydration. They also observed that mortars with broken PCM microcapsules experienced a delay in the hydration reaction compared to those with intact microcapsules, which was attributed to leakage of PCM into the mortar matrix and its potential interference with the surrounding material. Similarly, Hunger et al. [87] investigated the effects of MPCM microcapsules (Micronal DS 5008X) on the heat of hydration in self-compacting concrete. Through semi-adiabatic calorimetry, they found that incorporating MPCM in concrete reduced the hydration temperature and caused a delay in the hydration peak. In another study, the effects of MPCM on the hydration of cement mortars were examined using a semi-adiabatic method [157]. The results showed that increasing the PCM content in the mortar resulted in a significant reduction in the temperature rise during hydration, as well as a delay in the setting time. Moreover, higher PCM content was found to decrease the degree of hydration and contribute to a decrease in the rate of strength development compared to the reference mortar.

On the other hand, in an investigation on calcium sulphoaluminate (CSA) cement [158], where the water-to-cement ratio was 0.75, it was found that the inclusion of MPCM did not delay the heat of hydration. The hydration heat flow, as measured by isothermal calorimetry, for CSA with and without MPCM was almost the same for the first 3 days, indicating that MPCM did not significantly affect the heat of hydration in this case. Similarly, in the study on lightweight aggregate concrete (LWAC) incorporated with MPCM [159], the adiabatic temperature rise was examined. It was observed that, as shown in Fig. 6, the LWAC samples containing 5% and 10% MPCM showed a decrease in peak temperature by 3.1 °C and 4.3 °C, respectively, compared to the control group. Additionally, there was a delay in reaching the peak temperature by 2.9 and 5.1 h, respectively. These observations were attributed to the heat absorption capabilities of MPCM during liquefaction. The rate of adiabatic temperature rises in MPCM-LWAC slowed down near the phase transition temperature. Importantly, there were no significant differences in the characteristics of the adiabatic temperature rise curve, indicating that the presence of MPCM had no noticeable impact on cement hydration.

These findings suggest that the effect of MPCM on the heat of hydration and cement hydration process can vary depending on the specific cement type and composition. In the case of CSA cement and LWAC, MPCM did not significantly delay the heat of hydration or impact the cement hydration process. However, it is important to consider the specific characteristics of the cementitious system and MPCM used when evaluating the potential effects on heat generation and hydration kinetics. In summary, the author observed that many studies reporting a delay in the hydration reaction due to the addition of MPCM into cementitious composites have observed leakage of PCM into the composite matrix. These studies have predominantly used polymeric shell materials for microcapsules, which have a higher likelihood of collapsing during mixing. To overcome this issue and obtain more accurate results, it is suggested that further research should focus on using shell materials with improved mechanical performance compared to polymeric shells. By utilizing more robust shell materials, the potential for leakage and interference with the hydration process could be minimized, providing a better understanding of the effects of MPCM on cement hydration.

6.1.3. Mechanical and physical properties

The concept of incorporating MPCM in cementitious materials aims to improve thermal management in buildings and reduce their operational carbon footprint. However, it is crucial to ensure that this incorporation does not significantly affect the most important property of cementitious materials, which is compressive strength. Numerous research studies have demonstrated that the presence of MPCM in cementitious composites strongly influence the early-age mechanical strength development, typically up to 28 days. In general, most studies have concluded that the addition of MPCM to concrete or mortar leads to

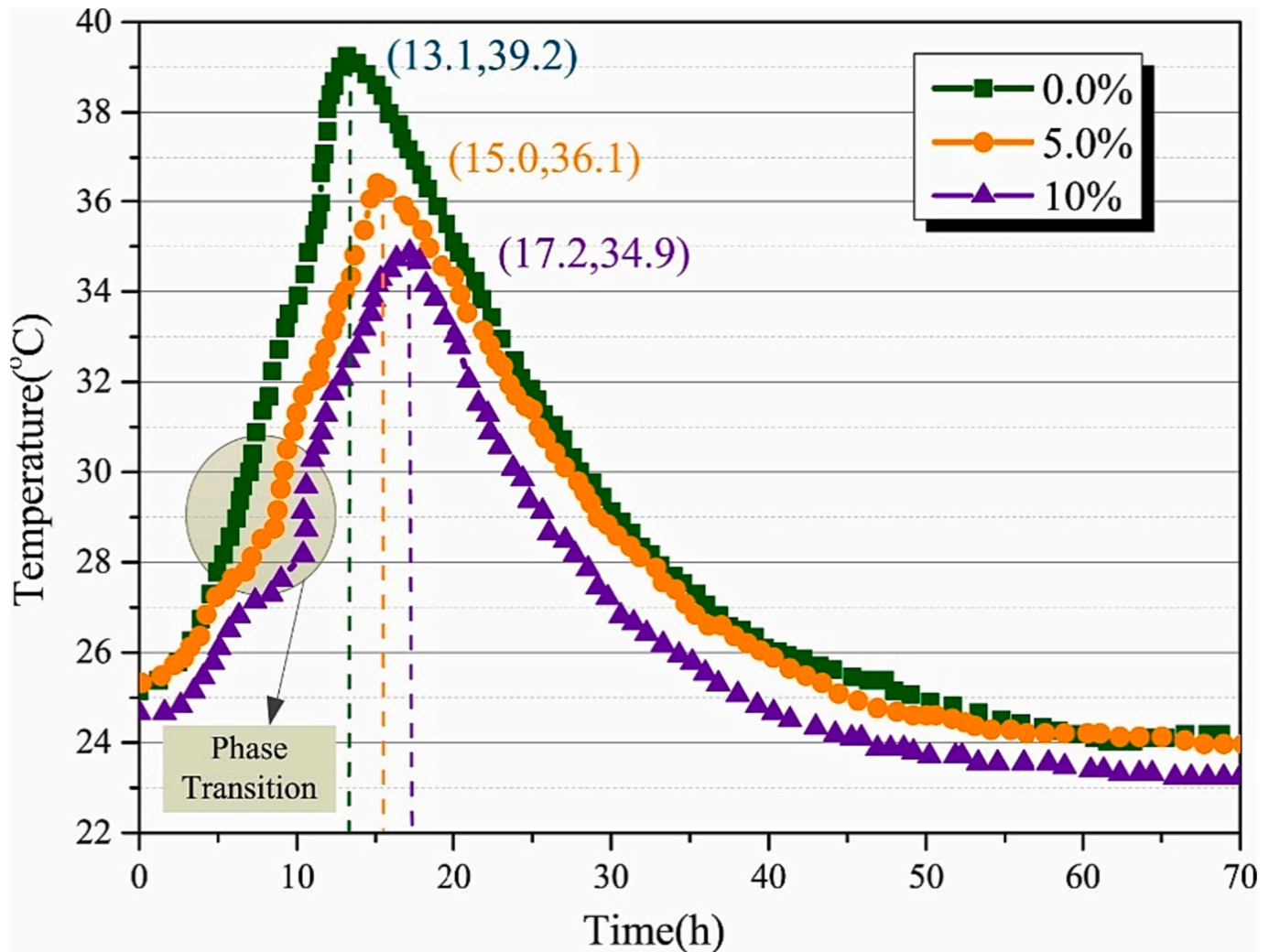


Fig. 6. Curves of adiabatic temperature rise of MPCM-LWAC [159].

a significant reduction in compressive and flexural strength. The literature highlights several common reasons for this strength reduction, as summarized in Fig. 7: weak bonding between microcapsules and the cement matrix, the soft inclusion effect of microcapsules, increased porosity, broken shells, leakage of melted PCM, and low strength of the microcapsules.

It is important to note that the type of shell material used for the microcapsules has a significant impact on the overall mechanical performance when incorporated into cementitious composites. Lecompte et al. [151] investigated the mechanical and thermophysical properties of concretes and cement mortars containing PCM microcapsules and compared their results with the modified Bolomey model. They concluded that MPCM can be considered as voids since they do not contribute to mechanical strength. However, a few studies in the literature have attributed the reduction in compressive strength to poor workability and the loss of free water available for the hydration reaction due to the inclusion of PCM microcapsules [86,100].

Indeed, there are studies in the literature that have reported improved compressive strength results when incorporating MPCM into cementitious composites. These studies often observed strength improvement at low dosage amounts of MPCM. Ismail et al. [12] conducted compressive strength tests on mortars embedded with silica-coated PCM microcapsules (SCPCM), as shown in Fig. 8. They found that at a lower dosage of 10% of SCPCM, the 28-day compressive strength of the cement mortar slightly improved by 2%. Although there

was a slight compressive strength loss at higher incorporation levels at all ages, the mortars remained strong enough for construction applications. The authors attributed this improved strength to the silica coating contributing to the formation of more C-S-H gel, which is responsible for strength development in cementitious materials.

In another study, a lightweight concrete was prepared by incorporating MPCM into foam concrete to enhance the thermal performance of buildings [160]. The researchers found that the inclusion of MPCM increased the bulk density and compressive strength of the foam concrete by up to 592.1 kg/m^3 and 2.52 MPa, respectively. This indicates that MPCM can contribute to both improved thermal properties and enhanced mechanical strength in certain cases. More also, it is worth noting that the incorporation of MPCM has been reported to lead to a reduction in the elastic modulus of concrete and a decrease in the density of the cementitious composite [161–163]. This reduction in density can make the cementitious composite suitable for lightweight structures, with an acceptable range of strength loss [162,164,165].

Recent studies on the compressive and flexural strength of cementitious composites after integrating MPCM are summarized in Table 10, showing a clear reduction in mechanical performance as the MPCM content increases. However, the reduction in compressive and flexural strength is more pronounced in studies that used organic (polymer) shell. This is because polymeric shells have low mechanical strength with their positions in the cement composite matrix acting as the weakest spots for stress to pass through under compression or flexure.

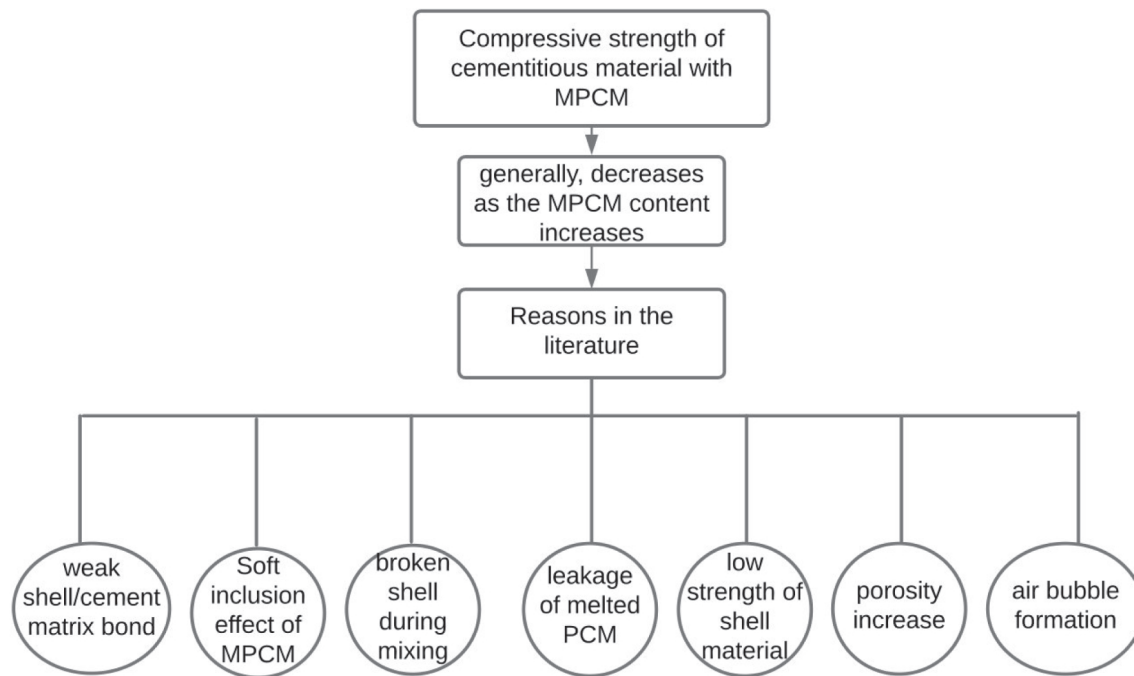


Fig. 7. Effect of MPCM on compressive strength of cementitious material.

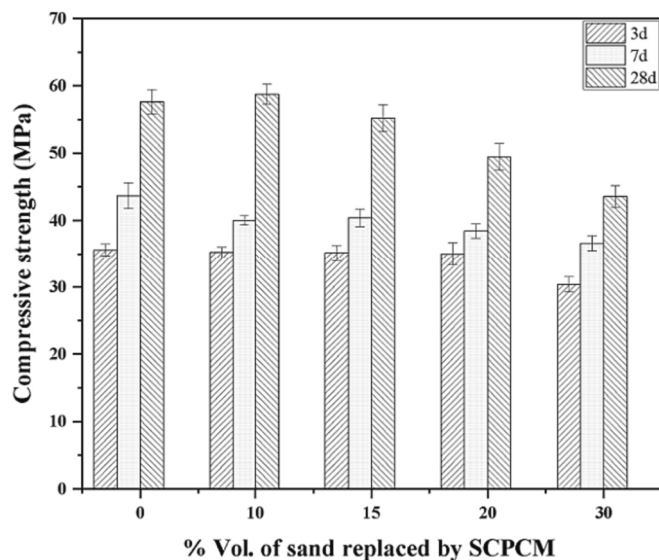


Fig. 8. Compressive strength of mortars incorporated with SCPCM [12].

Table 10 also shows that adding or replacing some sand with the MPCM really has no significant effect on the overall mechanical performance of the cementitious composite. Overall, exploring the potential of MPCM incorporation in cementitious materials offers exciting opportunities to enhance thermal performance of buildings while reducing operational carbon. However, careful consideration of the optimum dosage of MPCM is necessary to mitigate the significant drop in mechanical properties. This research can not only lead to practical solutions that meet structural requirements but also facilitate a deeper understanding of cementitious composites with MPCM, leveraging advanced techniques such as deep learning for predictive modeling during their service life.

6.1.4. Thermal properties

In order to enhance thermal performance of buildings and reduce heat loss through the building envelope, cementitious composites are integrated with MPCM. MPCM possess the unique ability to store and release heat in a latent form, making them suitable for improving the thermal properties of building materials. One important aspect to consider when incorporating MPCM into cementitious composites is the effect on thermal conductivity. Thermal conductivity is a measure of how quickly heat passes through a material, and it is a crucial factor in determining the overall thermal performance of a building. Numerous studies reported in the literature have consistently shown that the inclusion of MPCM in cementitious composites leads to a reduction in thermal conductivity.

As the amount of MPCM in the matrix increases, the thermal conductivity of the cementitious material decreases [149]. This decrease can be attributed to the relatively low thermal conductivity of the PCM and/or the shell material compared to the concrete or mortar matrix [16,176]. Ismail et al. [12] conducted a study comparing the thermal conductivity of reference cement mortar with mortars containing uncoated and silica-coated MPCM. The results showed that the addition of MPCM generally led to a decrease in the thermal conductivity of the mortar. This decrease was attributed to the lower thermal conductivity of the PCM compared to the sand it replaced. Specifically, when the mortar was replaced with 10% uncoated MPCM, the thermal conductivity decreased to 1.42 W/mK, which was 15% lower than that of the reference mortar (1.68 W/mK). Increasing the replacement level to 20% resulted in a 21% reduction in thermal conductivity. The silica-coated MPCM mortar exhibited slightly higher thermal conductivity than the uncoated MPCM mortar at all replacement levels, which was attributed to the silica coating. In another study involving gypsum composites, the thermal conductivity was investigated with the incorporation of MPCM at 5%, 10%, and 15% by weight [177]. The results showed that increasing the MPCM content in the composites resulted in a decrease in thermal conductivity. This was attributed to the lower thermal conductivity of the MPCM compared to the gypsum matrix. The composite specimen made solely of gypsum had the highest thermal conductivity at

Table 10
Compressive and flexural strength of cementitious composite with MPCM.

Cementitious composite	Shell type	Incorporation technique	% MPCM in mix	~ % 28d Compressive strength reduction*	% flexural strength reduction*	Ref.
Concrete	Organic (polymer)	Replacement (wt.)	1, 3, 5	33, 63, 69	–	[15]
Concrete	Organic (polymer)	Addition (vol)	1, 3, 5	14, 39, 47	–	[156]
Concrete	Organic (polymer)	Addition (vol)	8.6, 13, 25	44, 61, 85	–	[151]
concrete	Inorganic	Replacement (wt.)	10	34	–	[166]
Concrete	Organic (polymer)	Addition (wt.)	3.2	68	–	[162]
Micro concrete composite	Organic (polymer)	Addition (wt.)	0.025, 0.05, 0.075, 0.1	17, 20, 21, 39	–	[167]
Glass fiber concrete	Organic (polymer)	Addition (wt.)	5, 10, 15	21, 28, 38	3 (increase), 0, 21	[168]
Normal strength concrete	Inorganic	Addition (wt.)	1, 3, 5, 7 10	6, 15, 22, 29	–	[149]
High strength concrete	Inorganic	Addition (wt.)	1, 3, 5, 7 10	5, 13, 20, 26	–	[149]
3D printed concrete	Organic (polymer)	Addition (wt.)	10, 20	24, 44	4, 39	[145]
Shotcrete	Organic (polymer)	Addition (wt.)	10, 20	17, 36	–	[128]
Mortar	Inorganic	Replacement (wt.)	10,15,20,30	2(increase), 3, 14, 24	–	[12]
Mortar	Organic (polymer)	Addition (wt.)	5, 10, 15	31.2, 42.8, 50.4	–	[169]
Mortar	Organic (polymer)	Addition (wt.)	1, 3, 5	4, 5, 10	–	[170]
Mortar	Organic (polymer)	Replacement (wt.)	5, 10, 15, 20	14, 20, 37, 39	16, 36, 42, 47	[14]
Mortar	Organic (polymer)	Replacement (wt.)	5, 10	15, 54	–	[171]
Mortar	Organic (polymer)	Replacement (wt.)	10, 20, 30, 40	27, 49, 88, 92	–	[172]
Mortar	Organic (polymer)	Addition (wt.)	5, 10, 15, 20	65, 76, 83, 88	32, 56, 60, 67	[163]
Mortar	Organic (polymer)	Addition (wt.)	14.8, 22.1, 29.3	50, 62, 73	–	[151]
Mortar	Organic (polymer)	Replacement (vol.)	2.75, 5.5, 8.25, 11, 13.75, 16.5	4, 14, 20, 18, 31, 45	–	[173]
Mortar	Organic (polymer)	Addition (wt.)	5, 10, 15, 20	64, 76, 84, 88	32, 56, 60, 67	[163]
Polymer modified mortar	Organic (polymer)	Addition (wt.)	5, 10, 20	42, 65, 78	28, 35, 44	[174]
Cement paste	Organic (polymer)	Replacement (wt.)	10, 20, 30, 40	6, 48, 62, 86	–	[172]
Cement paste	Organic (polymer)	Replacement (vol.)	20, 40	35, 61	–	[175]
UHPC	Organic (polymer)	Addition (wt.)	5, 10	27, 35	24 (increase), 35	[100]
LWAC	Organic (polymer)	Replacement (wt.)	2.5, 5, 7, 10	8, 23, 29, 33	–	[159]

*The compressive strength and flexural strength shown in Table 10 are approximated to the nearest integer.

0.207 W/mK. However, when MPCM was incorporated at 5%, 10%, and 15%, the thermal conductivity decreased to 0.194 W/mK, 0.189 W/mK, and 0.183 W/mK, respectively.

Similar findings were reported by Feng Su et al. [178], where they investigated paraffin-based MPCM/gypsum composites for thermal energy-saving purposes. The study demonstrated that different weight ratios (10%, 20%, and 30%) of MPCM reduced thermal conductivity by up to 40%. Zhang et al. [179] also observed a decrease in the thermal conductivity of pure gypsum composites from 0.470 W/mK to 0.253 W/mK with the addition of 50% by weight of MPCM. Additionally, a study showed that even a small dosage of MPCM (up to 5%) reduced the thermal conductivity of concrete by about 37% [15], although very low dosages may not have a significant effect on the thermal conductivity of concrete [95]. Furthermore, a study measured the thermal conductivity of casted mortar and 3D printed concrete with 10% and 20% MPCM incorporation levels [145]. The results showed that the thermal conductivity of both cementitious composites decreased as the amount of MPCM increased. This reduction was attributed to the low thermal

conductivity of the PCM compared to the cementitious matrix and the interfacial gaps between MPCM particles and the matrix. Moreover, the thermal conductivity of the 3D printed composite with no MPCM was about 8% lower compared to the control sample of the cast composite, which was attributed to the interlayer interface of the strips in the 3D printed composite. The reduced thermal conductivity may also have advantages, as it can be utilized to improve the thermal insulation capacity of concrete [165].

Another reason for the reduced thermal conductivity of cementitious composites when MPCM are incorporated is the increase in air voids and reduction in density of the composite matrix due to the addition of microcapsules [175,180–182]. This consequently leads to an increase in thermal resistance within the material, resulting in lower thermal conductivity. To increase the thermal conductivity of cementitious composite integrated with MPCM, the air voids should be reduced. In a recent study by Lopez-Arias et al. [183], the vacuum encapsulation method was used to directly add PCM into hardened porous cementitious composites prepared at different water-to-cement ratios. The

results showed that the addition of PCM to mortars resulted in an increase in thermal conductivity. This increase was attributed to the PCM replacing the air voids that were present in the porous cementitious composite. Since air has a lower thermal conductivity compared to the paraffin wax PCM, the replacement of air with PCM led to an elevated thermal conductivity in the specimens. The potential problem with this approach is with almost all the air voids eliminated, such construction materials may not be suitable for cold climate regions with freeze–thaw cycle.

Another approach to improve the thermal conductivity of cementitious composites after the incorporation of MPCM is by adding conductive materials directly into the mortar to enhance heat transfer. Xiong et al. [184] added graphite platelets (GP) along with MPCM to cement paste. The addition of GP improved the thermal conductivity, heat storage coefficient, and heat storage/release rate of the paste. At a 5 wt% GP content, the thermal conductivity of the cement improved by 44.5–50.1%, and the storage/release rates increased by up to 25.3% and 23.8%, respectively. However, this approach could cause significant reduction in compressive strength of the cementitious composite since the conductive materials have low mechanical strength.

6.1.5. Frost resistance

Frost action is one of the leading causes of degradation in mortar and concrete [185]. In regions with cold climates, buildings are exposed to freezing temperatures, which can cause the pore solution within the concrete to freeze [186]. Since the volume of ice is larger than that of the pore water, the freezing process generates expansive pressure within the concrete if there is insufficient space to accommodate this expansion [187]. To mitigate the detrimental effects of freeze–thaw cycles in cementitious composites, the incorporation of MPCM with low transition temperatures has been proposed [188,189]. Pilehvar et al. [188] investigated the impact of freeze–thaw cycles on the mechanical behavior of geopolymer and Portland cement concrete incorporated with MPCM. They observed that concrete without MPCM experienced a significant reduction in compressive strength after 28 freeze–thaw cycles. In contrast, the concrete samples incorporated with MPCM exhibited less than 2.5% loss in compressive strength after the same number of freeze–thaw cycles. Tian et al. [190] conducted a numerical investigation on the thermal performance of composite concrete piers integrated with MPCM. Their findings indicated that compared to concrete piers without MPCM, the incorporation of MPCM resulted in a maximum reduction of up to 94% in freeze–thaw cycles, thereby extending the service life by 1470% at 0 °C. However, the durability of the MPCM was not specifically evaluated in their study.

Liu et al. [191] evaluated the frost resistance, internal pore structure,

and meso-deterioration of concrete during freeze–thaw cycles. They observed that the incorporation of MPCM significantly increased the frost life of the concrete and effectively impeded heat diffusion and temperature change. Their results also demonstrated that at the 100th freeze–thaw cycle, the rate of porosity change varied with different MPCM contents (0%, 5%, 10%, 20%): 41.60% > 28.98% > 17.70% > 9.39%, respectively. This indicates that during the freeze–thaw cycles, the MPCM consistently released heat during the freezing process and absorbed heat during the melting process, effectively enhancing the frost resistance of concrete. Additionally, CT images of the reference concrete and concrete modified with 10% MPCM, Fig. 9, showed a notable difference. The reference concrete exhibited numerous interface cracks between the aggregate and hardened cement paste, both at the edges and internally. However, in the concrete modified with 10% MPCM, minimal cracking was observed in these areas. The researchers concluded that the interface transition zone (ITZ) serves as the starting point for concrete damage and fracture occurrence. Furthermore, central CT images in Fig. 10 demonstrated that the internal deterioration of the reference concrete was the most rapid, with pores and cracks increasing significantly with freeze–thaw cycles.

The optimal amount of MPCM to be added to mortar or concrete for enhanced frost resistance is a critical consideration. Rodriguez et al. [192] conducted an in-depth analysis of the frost resistance of mortar with MPCM. They observed that when the volume replacement of paste by MPCM was 10%, the mortar exhibited efficient frost and scaling resistance. However, at 30% volume replacement, the control mortar without MPCM performed better in terms of frost resistance. Similarly, Liu et al. [191] found that when the MPCM content exceeded 10%, the pore structure of the concrete remained relatively stable during freeze–thaw cycles. They concluded that 10% MPCM content is the optimal amount for achieving improved frost resistance.

Further research, particularly in the context of buildings, is necessary to determine the optimum amount of MPCM to be incorporated into concrete. This is important as the addition of MPCM has the potential to mitigate the degradation of concrete buildings in cold climates. Additionally, it is crucial to investigate the freeze–thaw resistance of MPCM themselves, as freeze–thaw is a durability phenomenon that can impact the performance of MPCM in concrete. By conducting comprehensive studies, researchers can gain a better understanding of the optimum MPCM content and its impact on the frost resistance and durability of concrete structures.

6.1.6. Fire resistance

The flammability of organic PCM, particularly paraffins, poses a significant challenge to their widespread use in building materials.

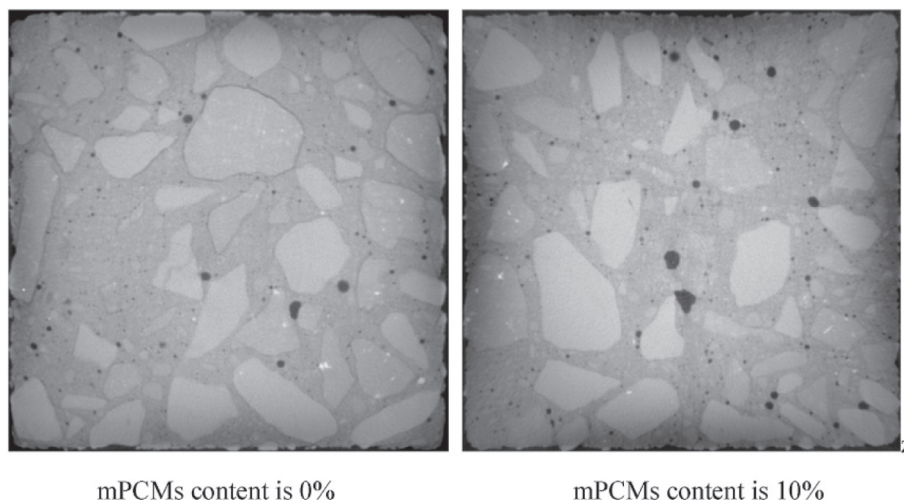


Fig. 9. Comparison of CT images at 175 freeze–thaw cycles (1/4 places) [191]

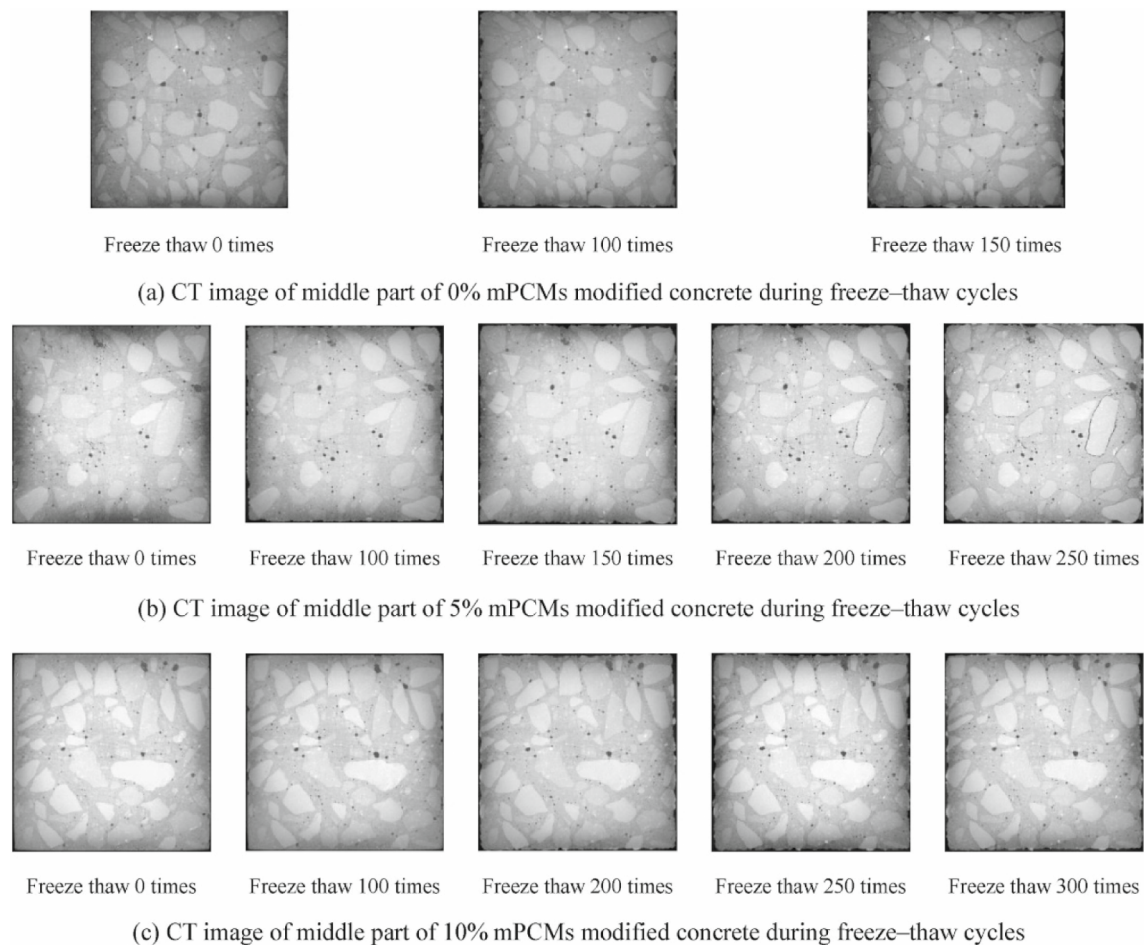


Fig. 10. Comparison of concrete deterioration during freeze–thaw cycles (Middle) [191].

Various parameters are used to assess and quantify the flammability of PCM, including flame spread velocity [193], flash point value, auto-ignition temperature, and heat release rate (HRR) [194]. It is expected that the flammability of cementitious composites increases with the amount of organic PCM. Haurie et al. [195] conducted dripping tests and smoke release tests to examine the fire behavior of mortars with different mass fractions of PCM. The dripping test results indicated that the reference mortar was not flammable, while the samples with 10% and 25% PCM exhibited an ignition time of approximately 38 s. The smoke test showed that at high temperatures, mortars with PCM released higher amounts of smoke, resulting in reduced light intensity and impaired visibility. The sample with 25% PCM caused a 40% reduction in light intensity compared to a 15% reduction observed with 10% PCM, indicating that smoke production increases with PCM content.

To address this limitation and unlock the full potential of PCM in cementitious materials, recent studies have explored the addition of flame retardants to the PCM or the shell materials. The incorporation of flame retardants can enhance the fire resistance of the composite. For example, Kazanci et al. [196] added fire retardants to PCM and encapsulated them in polystyrene microcapsules. They replaced 10% of their self-compacting concrete mixture with the MPCM. The results showed a 67% reduction in heat of combustion and a decrease in mass loss from 10.5% to 7.6% compared to the PCM without fire retardant. The fire retardant also improved the thermal stability and TES properties of the MPCM. Notably, with the addition of the fire retardant, the concrete block was classified as A2/B, d0, making it suitable for use in buildings for TES purposes according to EN 13501-1:2007 + A1:200. In another study, the fire endurance of concrete incorporated with lightweight

aggregates (LWA) infused with PCM and protected with a fire-resistant coating was investigated [197]. The coating prevented the leakage of PCM into the concrete, thereby preventing flammable reactions between the PCM and oxygen. However, the coating was effective only up to 250 °C, beyond which it burned and could no longer provide the necessary protection to the concrete. This study highlights the importance of encapsulation techniques in delaying the flammability reaction of PCM and preventing early leakage.

Similarly, Kang et al. [198] studied the flame-retardant performance of composites incorporated with capric acid, with or without halloysite nanotube PN component as a flame retardant in the methyl methacrylate shell. An additional layer of epoxy resin was applied to both MPCM to form a double coating. The results showed that the flame-retardant MPCM exhibited a 35.7% lower peak HRR, and 18.97% lower total heat release compared to MPCM without flame retardant. The flame retardant MPCM also demonstrated better thermal stability and lower temperature fluctuation in the composite.

The addition of flame retardants has shown to improve both the fire resistance and thermal performance of MPCM containing organic PCM in cementitious composites. However, it is important to note that many recent studies on MPCM in cementitious composites have not extensively discussed fire behavior. Further research on the thermal stability of MPCM in cementitious composites under fire conditions is necessary, as the leakage of PCM into the cementitious matrix in a fire hazard scenario may pose a serious threat.

6.2. Gypsum wall board incorporated with MPCM

In the United States, gypsum wall board, commonly known as

plasterboard or drywall, is the predominant indoor building material. With approximately 30 billion square feet of this material used annually in domestic construction, it plays a vital role in the building industry. Gypsum board is favored for its numerous advantages, including fire resistance, widespread availability, affordability, ease of installation, sound attenuation properties, and non-toxic nature [199]. Given the significance of gypsum board in construction, it is important to evaluate the potential impact of incorporating MPCM into this material.

6.2.1. Mechanical properties

Incorporating MPCM into gypsum wall board has been found to have a negative impact on the compressive and flexural strength of the material. Several studies in the literature have reported a reduction in strength with increasing amounts of MPCM, primarily attributing it to the incompatibility between the organic shell and the inorganic gypsum matrix. For instance, Zhang et al. [200] conducted compressive strength tests on gypsum material incorporated with different percentages of MPCM. They observed a decrease in compressive strength of the board, ranging from 24.7% with 3 wt% MPCM to 56.1% with 20 wt% MPCM. Similar findings were reported by another study [201], where the compressive and flexural strengths of gypsum composites decreased as the MPCM content increased. The reduction in strength was attributed to the water-repellent nature of the organic shell, resulting in weak interfacial bonding between the microcapsules and the gypsum matrix.

In another study investigating the mechanical performance of gypsum composites incorporated with MPCM [202], two types of boards were prepared: one with MPCM partially replacing the aggregates and another where only MPCM was used without the aggregate. The researchers observed a reduction in compressive and flexural strengths of the boards as the amount of MPCM increased. However, the study did not provide a clear reason for this general reduction in strengths. Additionally, a study by Srinivasaraonik et al. [203] investigated the mechanical properties of gypsum board incorporated with PCM microencapsulated in a melanin formaldehyde shell. They found that the compressive and flexural strengths decreased as the mass percentage of MPCM increased. Incorporating 5% and 10% MPCM resulted in reductions in compressive strength of 24.2% and 47.9%, respectively, and reductions in flexural strength of 22% and 31%, respectively. In this case, the decrease in strength was attributed to the lower apparent density of the composite caused by the addition of MPCM.

It is worth noting that most of the studies in the literature on gypsum wall boards primarily focus on their thermal performance after incorporating MPCM, as plaster boards are often not load-bearing walls. However, considering the mechanical properties becomes crucial if the replacement content percentage is high. It is important to determine the optimum replacement level of MPCM to ensure that the final product complies with relevant standards, such as ASTM Specification C 1396. Further research and development are needed to explore methods to mitigate the reduction in mechanical strength caused by the incorporation of MPCM into gypsum board. Strategies such as improving the compatibility between MPCM and the gypsum matrix or optimizing the encapsulation techniques may help overcome this limitation. By finding a balance between thermal performance and mechanical strength, it may be possible to achieve gypsum board formulations that effectively incorporate MPCM while maintaining the required strength for structural integrity and meeting industry standards.

6.2.2. Thermal conductivity and heat capacity

As mentioned previously, studies in the literature have primarily focused on the thermal enhancement of wallboards through the incorporation of MPCM. Both numerical simulations and experimental investigations have consistently demonstrated that the addition of MPCM to drywalls results in a reduction in thermal conductivity and a significant increase in heat capacity compared to conventional wallboards [91,117,204–206]. A recent study showed that the thermal conductivity of gypsum board with 10% MPCM reduced significantly, and this was

attributed to the polymeric material of the shell encapsulating the PCM [207]. Another reason stated in the literature for the reduced thermal conductivity of MPCM integrated wallboards is the PCM lower thermal conductivity [91]. Furthermore, the heat capacity of plaster boards with MPCM increases with the level of replacement until an optimum amount is reached, beyond which further replacements lead to a reduction in heat capacity [91,208].

It is worth mentioning that the optimal amount of MPCM to be added to the wallboard matrix may vary in different research studies. This discrepancy could be attributed to variations in experimental conditions, properties of the PCM used, material properties of the microcapsules, characteristics of the wallboard, and the method of preparation of the MPCM-incorporated wallboards. Several recent studies have investigated the effects of MPCM on the thermal properties of wallboards, and their findings are summarized in Table 11. These studies provide further evidence that the incorporation of MPCM enhances the thermal performance of wallboards, offering significant potential for improved thermal management in buildings.

6.2.3. Fire resistance

Gypsum is primarily composed of water that has been chemically mixed, which significantly enhances its efficiency as a fire-resistant barrier. When gypsum drywall is burned, the water is gradually released as steam, which slows the transfer of heat. Using this gypsum, drywall that resists fire is created and referred to as Type X designed to meet fire performance requirements stated in the ASTM C1396 [216]. In a bid to enhance the thermal performance of ordinary drywalls, MPCM can be added to the wallboard matrix. However, the caveat is that study has shown that the addition of paraffin PCM can increase the flammability of wallboards compared to plain wallboards [217]. Microencapsulation can help contain the PCM and prevent it from permeating the porous structure of the drywall. Nevertheless, in the event of a fire, where temperatures can reach approximately 800 °C [218], there is a possibility that the microcapsules may fail, leading to PCM leakage into the plaster board's porous structure and acting as fuel [219].

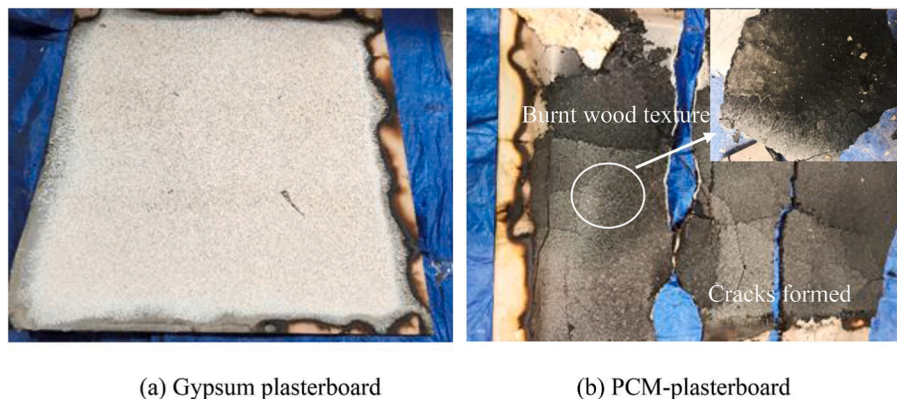
The fire resistance of PCM plasterboards and regular plasterboards was thoroughly analyzed by Gnanachelvam et al. [220]. In this inquiry, wallboards were tested for their thermal properties and subjected to small-scale, routine fire tests. The findings of thermal property tests revealed that the mass loss of PCM-plasterboard, which is caused by dehydration and evaporation, is very high at 40% at high temperatures, as opposed to the mass loss of gypsum plasterboard, which is caused by dehydration at a rate of 23%. The PCM-fire plasterboard's resistance was decreased by the extra fuel load that the evaporating PCM discharged during the fire. They observed the fire side of the PCM-plasterboard and the gypsum plasterboard. The board surfaces following the tests are as shown in Fig. 11. The paper surface of the gypsum plasterboard was burned and turned light grey, while the PCM-plasterboard was burned and had gone black, like the texture of burned or charred wood. As shown in Fig. 11(a), the plasterboard maintained its geometry without any spalling or cracking, but the PCM-plasterboard lost its integrity and developed significant cracking (b). Similarly, the fire performance of a commercially available PCM enhanced plasterboard was investigated [221]. They discovered that while the PCM plaster boards behave like charring solids and have the capacity to add a large amount of fuel to a compartment fire, they maintain their structural integrity throughout the flaming period. Additionally, they noticed that the orientation had little to no impact on the mass loss rate but increased the peak heat release rate by up to 25%.

To improve the fire resistance of MPCM, recent research has explored the incorporation of fire retardants along with PCM, leading to some promising results [196]. Most of the studies investigating the flammability of plaster boards enhanced with MPCM have utilized paraffins as the PCM. Paraffins are commonly used due to their commercial availability and reliable phase change properties. However, one significant drawback of paraffins is their flammability. It may be worthwhile to

Table 11

Studies on effect of MPCM on thermal properties of gypsum wallboard, main findings, and prospects.

Building material	Shell material	Shell material	PCM type	Key findings	Ref.
Gypsum composite	Melanin formaldehyde	Melanin formaldehyde	capric acid/palmitic acid	• The thermal conductivity of the gypsum composites diminishes as the amount of MPCM is increased. • By delaying the flow of heat from the outside to the inner surface, this decrease in thermal conductivity increases thermal comfort.	[203]
Gypsum composite	Silica	Silica	paraffins	With the weight ratio of MPCM at 10 wt%, gypsum materials were able to achieve high mechanical strength, compactness, thermal conductivity, and an excellent thermal regulation performance.	[209]
Gypsum wallboard	–	–	paraffin	• The adoption of wallboards with PCM-incorporated technology decreased the building's overall cooling demand by 1.1 and 1.5 percent, respectively. • Gypsum wallboard impregnation based on PCM decreased energy use and improved indoor thermal comfort.	[210]
Gypsum wallboard	–	–	–	For indoor thermal comfort and energy savings, hybrid PCM wallboard can function in a wide range of phase-change temperatures and can be used in a variety of climates. The hybrid PCM wallboard box uses over 30% less energy than gypsum wallboard.	[211]
Gypsum wallboard	Calcium alginate	Calcium alginate	RT25	• When PCM capsules are inserted, the thermal response of the gypsum model house to the fluctuating ambient temperature is slowed. • Increasing PCM capsule filling ratios improve the gypsum model house's ability to regulate temperature.	[212]
Gypsum board	–	–	n-octadecane	• When compared to standard gypsum board, functional gypsum board (FGB) had a 15% lower thermal conductivity.	[213]
Gypsum plaster mixture	Acrylic polymer	Acrylic polymer	Micronal DS 5008 X	• Fire tests revealed that the samples responded to the fire class, and adding PCM to gypsum plasters reduces their mechanical qualities.	[202]
Gypsum plasterboard	Acrylic polymer	Acrylic polymer	Micronal DS 5040X	• Gypsum plasterboard's thermal conductivity was decreased to 0.139 W/mK by adding 15% PCM. • With a temperature difference of about 1.5 °C, the MPCM gypsum plasterboard operated for a longer period of time than ordinary gypsum plasterboard. • Due to the addition of MPCM, the MPCM pasteboard had 0.4 W/min more energy stored than gypsum plasterboard.	[214]
Gypsum powder	Silica	Silica	paraffin	• When compared to gypsum wallboards, PCM wallboards showed higher latent heat and specific heat capacities. Data analysis showed a linear link between temperature and PCM wallboards' specific heat capacity.	[215]

**Fig. 11.** Fire sideboard surfaces after tests.

explore non-paraffin or inorganic PCM as alternatives in order to mitigate flammability risks. In a recent study, in order to reduce the flammability impact of paraffin PCM in wallboards, an extra layer of gypsum board was sandwiched in between two layers of PCM plasterboards [207]. This was found to enhance the fire resistance of the multi-layer board.

7. Future outlook

Microencapsulation is a promising method to directly integrate PCM into construction materials for improved thermal energy efficiency. Research on how to optimally utilize this great technology will be highly demanding in the coming years; especially, with the unprecedented global interest of operational and embodied carbon reduction. The following are the possible future outlooks for MPCM technology:

1. Considering the cost advantage and non-flammability of inorganic PCM, future research will be on addressing its phase separation and supercooling issues. While adding nucleating agents to inorganic PCM can be an effective short-term solution, there is a risk of phase separation occurring after multiple thermal cycles, which can ultimately compromise the performance and reliability of the PCM.
2. Regions with wide temperature range, particularly where precise temperature control and frequent thermal cycling are required, can benefit from the double endothermic peaks of binary eutectic PCMs. The main advantage of having a tunable PCM can make TES application in buildings available year-round. Future experimental studies will focus on optimizing the design application needs for single or tunable peak in regions with wide temperature profiles.
3. To fully harness the thermo-mechanical advantage of inorganic shells over organic shells, the future outlook will focus on naturally existing micron size hollow materials such as cenospheres. The

complex procedures and cost of production of MPCM can be reduced using these materials and better mechanical performance of construction materials with MPCM can be achieved. More also, most of the studies in the literature with hybrid shells encapsulate organic PCM. Due to the cost effectiveness of inorganic PCM, further research will be on encapsulating inorganic PCM within hybrid shells This can create economical MPCM that may be widely used.

4. The optimal amount of MPCM to be added to mortar or concrete for enhanced frost resistance is a critical consideration. Future research will need to investigate the long-term freeze–thaw resistance of MPCM as freeze–thaw is a durability phenomenon that can impact the performance of MPCM in cementitious composite.
5. Further research on the thermal stability of MPCM in cementitious composites under fire conditions is necessary, as the leakage of PCM into the cementitious matrix in a fire hazard scenario may pose a serious threat. Additionally, Further studies should investigate the coating of microcapsules with fire retardants, aiming to enhance the fire resistance of MPCM without significantly affecting their thermal conductivity. By addressing these fire-related concerns and exploring alternative PCM options, it is possible to develop MPCM-enhanced wallboards that not only exhibit improved thermal performance but also maintain a high level of fire safety.
6. It is worth noting that most of the studies in the literature on gypsum wall boards primarily focus on their thermal performance after incorporating MPCM, as plaster boards are often not load-bearing walls. However, considering the mechanical properties becomes crucial if the replacement content percentage is high. Further research and development are needed to explore methods to mitigate the reduction in mechanical strength caused by the incorporation of MPCM into gypsum board. Strategies such as improving the compatibility between MPCM and the gypsum matrix or optimizing the encapsulation techniques may help overcome this limitation.
7. There is paucity of information in the literature on the carbon emission reduction potential of MPCM and the economic advantage of this technology in construction materials. Critical analysis of the aforementioned can attract huge audiences, policy makers, and relevant stakeholders to the significance of MPCM for TES in building materials.

8. Conclusions

In this review, the potentials of microencapsulation as the preferable method of incorporation of PCM into building materials were critically evaluated. In this review study, general overview of PCM, encapsulation methods, shell material, thermal enhancement methods, and the influence of MPCM on mechanical and thermal properties of cementitious composites and wallboards as recorded in the literature were discussed and analyzed. The following conclusions can be drawn:

1. PCM can enhance building thermal management due to its ability to store/release heat in latent form. While most of the studies in the literature and commercially available PCM are paraffins produced from non-renewable petroleum resources, biobased PCM are safe for the environment and have a number of useful qualities, including high latent heats of fusion, little supercooling, robust thermal cycling, reasonable cost, and little volume change during phase transition. This PCM can be an alternative replacement for paraffins.
2. Generally, microencapsulation improves the thermal performance of PCM and prevent the leakage of melted PCM into construction material matrix. However, it also leads to reduction in enthalpy of PCM attributable to the low melting point of the microcapsules and/or small amount of core material within these shells. Most organic shells for PCM studied in the literature starts to degrade at temperature ≥ 200 °C and considering their flammability they can occlude large scale application MPCM in construction materials. Hybrid polymer shell materials on the other hand often times creates multiple exothermic peaks during crystallization. Agglomeration of inorganic shell materials are also observed in developed inorganic microcapsules which can prevent even distribution of MPCMs in construction materials.
3. Microcapsules made from polymer material exhibit low mechanical performance which are susceptible to collapse during mixing. Inorganic microcapsules exhibit better mechanical performance than organic microcapsules, but the shell materials of inorganic microcapsules may separate from the PCM during thermal cycling. Nonetheless, silica is the most widely used inorganic shell for PCM studied in the literature due to the compatibility and possible bonding between the silica and calcium hydrate during cement hydration, to form additional calcium silicate hydrate (C-S-H) gel. Inorganic shells compensate for the significant reduction in mechanical properties often observed in construction materials incorporated with MPCM. This advancement holds great potential for widespread application of MPCM in buildings, enabling improved thermal management and a reduction in operational carbon emissions.
4. Conductive nanomaterials are either added to the PCM or doped with the shell material to enhance the low thermal conductivity of PCM and/or microcapsules. In the literature, carbon nanotubes (CNTs) are frequently utilized as thermal conductivity enhancers for shell materials and PCM. This is because CNTs have excellent thermal conductivity and tensile strength and their nanoscale particle size make them easily blend with the PCM or shell material. Addition of these thermal enhancement nanoparticles into PCM lead to the reduction in latent heat of the PCM at higher wt.% incorporation level. Therefore, it is crucial to determine the optimum amount of these particles that will offer the best PCM thermal performance.
5. The incorporation of MPCM into cementitious composite have significant influence on their fresh properties. Workability of cementitious composite decrease with increasing amount of MPCM. The reasons as mentioned in the literature are the large surface area of the microcapsules, adsorption susceptibility of the shell material, morphology of the surface of the microcapsule, properties of the shell material, and leakage of the PCM into the matrix. There are conflicting reports in the literature regarding the effect of MPCM on cement hydration and the resulting heat of hydration. Some studies have shown that the inclusion of MPCM in cementitious composites delayed and reduced the heat of hydration, other studies reported that it has no significant effect on heat of hydration. However, the author observed that many studies reporting a delay in the hydration reaction due to the addition of MPCM into cementitious composites have observed leakage of PCM into the composite matrix. These studies have predominantly used polymeric shell materials for microcapsules, which have a higher likelihood of collapsing during mixing.
6. The shell material used for the microcapsules has a significant impact on the overall mechanical performance of cementitious composite. Mostly, the compressive and flexural strength of construction materials decrease with increasing amount of MPCM. However, this reduction is often higher when organic materials are used as microcapsules because of their relatively low mechanical property compared to the cementitious composite matrix or the sand they replaced.
7. Generally, the incorporation of MPCM in construction materials reduced their thermal conductivity. This is due to the low thermal conductivity of the PCM and/or the shell material compared to the construction material matrix or increase in air voids and reduction in density of the composite matrix due to the addition of microcapsules. To improve the thermal performance of the MPCMs, conductive nanoparticles are added either to the core material or the shell before incorporating them into the construction material. The main challenge with this approach is the reduction in latent heat storage of the MPCM. Attempt has also been made to significantly reduce the air voids in cementitious composite but the potential problem with this

approach is with almost all the air voids eliminated, such construction materials may not be suitable for cold climate regions with freeze–thaw cycle. Another method to enhance heat transfer is to add the conductive nanomaterial directly into the construction material. However, this approach could cause significant reduction in compressive strength of the cementitious composite since the conductive materials have low mechanical strength compared to the cementitious matrix.

8. Most of the studies that investigated wallboards with MPCM used either paraffin as core material and/or polymer materials as microcapsule. However, one of the major limitations of paraffin/polymeric shell is flammability. This drawback can prevent the industrial application of such MPCM wallboards. Microencapsulation prevents the leakage of PCM into cementitious matrix, thereby delaying the flammable reactions between the PCM and oxygen. The addition of flame retardants has shown to improve both the fire resistance and thermal performance of MPCM containing organic PCM in cementitious composites. However, it is important to note that many recent studies on MPCM in cementitious composites have not extensively discussed their fire behavior.

CRediT authorship contribution statement

Abdulmalik Ismail: Visualization, Validation, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization, Supervision, Writing – original draft, Writing – review & editing. **Jialai Wang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Babatunde Abiodun Salami:** Project administration, Methodology, Investigation, Formal analysis, Supervision, Writing – original draft, Writing – review & editing. **Lukumon O. Oyedele:** Methodology, Visualization. **Ganiyu K. Otukogbe:** Methodology, Visualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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